

On the Descent of Grossencharacters Arising from Jacobi Sums

James Gass

An essay submitted in partial fulfillment of
the requirements for the degree of
B.Sc. (Honours)

Pure Mathematics
University of Sydney



November, 2025

Abstract

This thesis investigates the connection between character sums, point counting on algebraic varieties, and Hecke Grossencharacters. An alternate proof is given of Weil's theorem that Jacobi sums, a certain type of character sum arising from point counts, are Grossencharacters, arguing through Anderson's formalism and related modern approaches. Then, a proof that Jacobi sums descend to Grossencharacters on their field of moduli is also presented, consequentially extending the class of primes at which they can be evaluated. We apply the notion of descent to investigate whether finite field hypergeometric functions, arising from more involved point counts, also descend.

CONTENTS

Chapter 1. Introduction	1
1.1. History and Motivation	1
1.2. Main Results	2
1.3. Applications	4
1.4. Contents	4
Chapter 2. Preliminaries	5
2.1. A brief introduction to algebraic number theory	5
2.2. Ideas from the Étale Cohomology and the Motivic World	11
Chapter 3. Jacobi Sums and Grossencharacters	13
3.1. Definitions	13
3.2. Gauss and Jacobi Sums	16
3.3. Reframing Jacobi Sums	17
3.4. Jacobi Sums are Hecke Grossencharacters	21
3.5. An application of Weil's theorem	24
Chapter 4. Descent	29
4.1. Definitions and motivations	29
4.2. Galois actions on J_θ	30
4.3. Descent of Jacobi sums and Hecke Grossencharacters	34
4.4. A view of the adelic framework	40
Chapter 5. Finite Field Hypergeometric Functions	42
5.1. Bridging the classical and finite field hypergeometric functions	42
5.2. Jacobi sums, descent, and hypergeometric functions	45
References	48

CHAPTER 1

Introduction

“And though the holes were rather small / They had to count them all / Now they know how many holes it takes to fill the Albert Hall.”

- Lennon & McCartney, *A Day in the Life*

1.1. History and Motivation

From Gauss’s pioneering work on quadratic and cubic reciprocity [10, p. 104, 440-445] to the Weil conjectures and beyond, the study of the number of solutions to polynomial equations modulo a prime number p has remained a central pursuit in arithmetic and number theory. One of the earliest yet significant examples appeared in Fermat’s investigation on determining which primes p gave rise to solutions to the equation $x^2 + 1 \equiv 0 \pmod{p}$. This was proven to be the case if and only if $p \equiv 1 \pmod{4}$, which consequentially allowed Fermat to show that a prime p can be written as the sum of two squares under the same condition [32, p. 3]. The equivalent notion of the polynomial $f(x) = x^2 - a$ splitting into linear factors modulo p was characterised by Gauss’ numerous proofs of the law of quadratic reciprocity, with the splitting of $f(x)$ merely depending on the congruence class of $p \pmod{4a}$ [32, p. 5]. This observation by Gauss revealed a much deeper structure, namely that arithmetic properties can be studied through periodic functions on the integers. In this case, the function in question is the quadratic character $\chi_a(p) = (a/p)$, a Dirichlet character given by

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } x^2 - a \text{ splits modulo } p; \\ -1 & \text{otherwise,} \end{cases}$$

and $\chi_a(p) = 0$ if $p \mid a$.

Whilst an effective tool for computing these manageable point counts, more complicated expressions often require more advanced machinery. For example, given the equation $X_p : y^2 = x^3 + 1 \pmod{p}$, where one can compute that the number of points satisfying X_p is given by

$\#X_p = p + \psi + \overline{\psi}$, where ψ is a complex number satisfying $\psi\overline{\psi} = p$. By enforcing certain conditions on ψ [13, p. 106], it can be shown that $\psi + \overline{\psi} = A \in \mathbb{Z}$ with the conditions $4p = A^2 + 27B^2$ and $A \equiv 1 \pmod{3}$ fixing the sign of A . Instead of arising from a Dirichlet character, ψ comes from a much broader class of characters, that of Grossencharacters on the ideals of $\mathbb{Z}[\zeta_3]$, which generalise Dirichlet characters by attaching values to the primes above p in the relevant number field. The aim of this thesis is to explore the character sums arising from counting points on Fermat hypersurfaces $b_1x_1^m + \cdots + b_rx_r^m \equiv 0 \pmod{\mathfrak{p}}$, and the Grossencharacters arising from their analysis.

1.2. Main Results

The first part of this thesis is spent recalling the setup to André Weil's paper [31] through the conventions of Mark Watkins's work [30], introducing Gauss and Jacobi sums. The latter is a character sum naturally arising from counting the number of points on a Fermat hypersurface modulo a prime ideal $\mathfrak{p}$, and is traditionally given by:

$$J_{\underline{a}}(\mathfrak{p}) := (-1)^{r+1} \sum_{x_1, \dots, x_r} \prod_{i=1}^r \chi_{\mathfrak{p}}(x_i)^{a_i}$$

where the sum above is taken over $x_1, \dots, x_r \in \mathbb{F}_{\mathfrak{p}}$ satisfying $1 + \sum_{i=1}^r x_i \equiv 0 \pmod{\mathfrak{p}}$, and $\chi_{\mathfrak{p}} : \mathbb{F}_{\mathfrak{p}} \rightarrow \mathbb{C}^1$ is the generating character modulo $\mathfrak{p}$. Gauss sums, on the other hand, are defined as

$$g_{a/m}(\mathfrak{p}) := \sum_{x \in \mathbb{F}_{\mathfrak{p}}} \chi_{\mathfrak{p}}(x) \Theta(x)$$

where $\Theta : \mathbb{F}_{\mathfrak{p}} \rightarrow \mathbb{C}^1$ is an additive character. The versatility of Gauss sums, already apparent in Gauss's proofs of quadratic reciprocity, becomes even clearer when we reinterpret the Jacobi sum as a product of Gauss sums:

$$J_{\underline{a}}(\mathfrak{p}) = \frac{(-1)^{r+1}}{q} g_{s/m}(\mathfrak{p}) \prod_{i=1}^r g_{a_i/m}(\mathfrak{p})$$

where $s = -\sum_i a_i$ (see Proposition 3.3.3 for further details).

This prompted us to re-frame Jacobi sums with the modern language of Anderson's formalism [1], closely resembling the setup in Watkins' paper [30] and its implementation in Magma [4]. The idea is to define Jacobi sums as a product of Gauss sums. Given $\theta = \{y_i\}_{1 \leq i \leq r}$ the multiset of

defining data of a Jacobi sum, subject to the condition that $\sum_i y_i \in \mathbb{Z}$ (see Definition 3.3.6 for more information regarding this restriction), we then define the notion of a formal Jacobi sum

$$J_\theta(\mathfrak{p}) := \prod_i g_{y_i}(\mathfrak{p})^{n_i}.$$

This leads to the first main result of the thesis due to Weil [31, p. 489] and proven by Anderson [1] with expository work by Watkins [30] in the following form.

Theorem 1.2.1. *Let $y_i = y'_i/m_i \in \mathbb{Q}$ be expressed in lowest terms and $m = \text{lcm}(m_1, \dots, m_r)$. Then $J_\theta : I_{\mathbb{Q}(\zeta_m)}((m^2)) \rightarrow \mathbb{C}^\times$ is a Hecke Grossencharacter.*

The proof of the above theorem relies heavily on the Stickelberger element and Stickelberger's theorem. Namely, given $\theta_a := \sum_{t \pmod{m}} \{ta/m\} \sigma_{-t}^{-1} \in \mathbb{Q}[\text{Gal}(\mathbb{Q}(\zeta_m)|\mathbb{Q})]$, we obtain a prime ideal decomposition of Gauss sums given by $(g_{y_i}(\mathfrak{p}))^m = \mathfrak{p}^{m\theta_{y_i}}$. Hence, Jacobi sums have the following ideal decomposition:

$$(J_\theta(\mathfrak{p})) = \mathfrak{p}^{\kappa(\theta)}$$

with $\kappa(\theta) := \sum_i n_i \theta_{y_i}$. Thus, we obtain from this the correct infinity type of the Grossencharacter up to a unit $u(\alpha)$. It is then shown that $u(\alpha) = 1$ by choosing a “large” enough defining ideal $\mathfrak{f}$.

Now equipped with Weil's result, we ask ourselves whether or not these Grossencharacters, defined over some base field L , descend to a Hecke character on their field of moduli $K \subseteq L$. Indeed, the main result of chapter 4 answers this question on Grossencharacters coming from Jacobi sums.

Theorem 1.2.2. *Let $\theta = \{a_i/m\}_{1 \leq i \leq r}$ be the defining data of the Jacobi sum J_θ , and $H = \{\sigma \in \text{Gal}(\mathbb{Q}(\zeta_m) | \mathbb{Q}) : J_\theta^\sigma(\mathfrak{a}) = J_\theta(\sigma(\mathfrak{a})) = J_\theta(\mathfrak{a})\}$. Then the Grossencharacter $J_\theta : I_{\mathbb{Q}(\zeta_m)}(\mathfrak{f}) \rightarrow \mathbb{C}^\times$ descends to a multiplicative character $J_{0,\theta}$ on its field of moduli $K = \mathbb{Q}(\zeta_m)^H$, and preserves the property of being a Grossencharacter.*

The proof of this theorem observed by Watkins [30, p. 4] follows the process of defining a new character $J_{0,\theta}$ on K such that, for all primes $\mathfrak{p} \in I_K(\mathfrak{f}_0)$ and $\mathfrak{P} \in I_{\mathbb{Q}(\zeta_m)}(\mathfrak{f})$ above $\mathfrak{p}$, $J_{0,\theta}(\mathfrak{p}\mathcal{O}_K) = J_\theta(\mathfrak{P}\mathbb{Z}[\zeta_m])$. After showing that this choice of character is well-defined, an application of Weil's theorem yields the desired result. One of the main takeaways from this theorem is that descended Grossencharacters can be evaluated at more classes of primes (for example, primes $\mathfrak{p}$ such that $\text{Nm}(\mathfrak{p}) \not\equiv 1 \pmod{m}$). See Example 4.3.6 to see this in action.

1.3. Applications

Finite field hypergeometric functions (HGMs), tracing their origins back to John Greene’s paper [11], form a natural setting to discuss the applications of the main theorems of this thesis. We discuss the following question pertaining to the study of HGMs.

Question 1.3.1. Do finite field hypergeometric functions descend?

To define an HGM, the requirement that $\alpha_i, \beta_j \in \frac{1}{q-1}\mathbb{Z}$, for defining data α, β , $\alpha_i \in \alpha$ and $\beta_j \in \beta$, and $q = \text{Nm}(\mathfrak{p})$, means that we are restricted to computing HGMs at primes satisfying $\text{Nm}(\mathfrak{p}) \equiv 1 \pmod{m}$. Here, m is the lowest common multiple of the denominators of the data α and β . Having expressed HGMs (per [14]) as sums of Grossencharacters coming from Jacobi sums, we turn to the question of whether they too descend, and if it meaningfully preserves their fundamental property of counting points. Frits Beukers, Henri Cohen and Anton Mellit gave a proof of a version of this result for HGMs defined over $\mathbb{Q}$, meaning that their data α, β gives polynomials $\prod_r (x - e^{2\pi i \alpha_r})$, $\prod_s (x - e^{2\pi i \beta_s})$ with coefficients in $\mathbb{Z}$. Thus, when HGMs arise from more intricate point counts, descent provides a means to evaluate them over a wider class of primes.

1.4. Contents

Chapter 3 begins with an exploration of the rich interplay between Jacobi sums, arising from point counts of Fermat hypersurfaces, and Hecke Grossencharacters. After exploring the proof due to André Weil connecting these two concepts, we then investigate the notion of (Galois) descent (chapter 4). This is a concept which can trace its roots back to the advent of complex numbers. For example, one can then identify the real numbers $\mathbb{R}$ with the subset of $\mathbb{C}$, $\{\alpha \in \mathbb{C} : \bar{\alpha} = \alpha\}$ of elements invariant under complex conjugation. The goal of chapter 4 is to explore the descent of Hecke characters, culminating in a proof that the Grossencharacters given by Jacobi sums descend to their field of moduli. We finalise this chapter with a refinement of this argument by viewing Grossencharacters through the adelic framework (4.4.1). Chapter 5 concludes the expository work of this thesis by exploring the applications of descent to “higher-order” character sums, such as finite field hypergeometric functions (HGMs). We ask ourselves if HGMs descend to their field of moduli, thus alleviating the restrictions on the classes of primes which they can be computed at.

CHAPTER 2

Preliminaries

2.1. A brief introduction to algebraic number theory

In this chapter, we refer to a finite extension K of $\mathbb{Q}$ as a number field. Moreover, given a ring A , we recall that $\text{Spec}(A)$ is the set of prime ideals.

2.1.1. Rings of integers, factorisations, and the class group. We begin by recalling the definition of a Dedekind domain.

Definition 2.1.1. *A Dedekind domain $\mathcal{O}$ is an integral domain satisfying the following properties:*

- a) $\mathcal{O}$ is a Noetherian ring;*
- b) $\mathcal{O}$ is integrally closed;*
- c) Every $(0) \neq \mathfrak{p} \in \text{Spec}(\mathcal{O})$ is a maximal ideal.*

Equipped with the notion of a Dedekind domain, we wish to study what is known as the “ring of integers” $\mathcal{O}_K$. Recall that α is an algebraic integer if it is the zero of a monic polynomial $f \in \mathbb{Z}[x]$.

Definition 2.1.2. *We denote the ring of integers $\mathcal{O}_K$ of a number field K as the set:*

$$\mathcal{O}_K = \{\alpha \in K : f(\alpha) = 0 \text{ for } f \in \mathbb{Z}[x] \text{ monic}\}.$$

Indeed, the ring of integers of a number field is a Dedekind domain, which is demonstrated by arguing that, for $0 \neq \mathfrak{p} \in \text{Spec}(\mathcal{O}_K)$, $\mathcal{O}_K/\mathfrak{p}$ is a finite set and an integral domain, thus making it a field. A classic example of the ring of integers is the case where $K = \mathbb{Q}(\sqrt{d})$, for $d \in \mathbb{Z}$ square-free. In this case, $\mathcal{O}_{\mathbb{Q}(\sqrt{d})} = \mathbb{Z}[\sqrt{d}]$ if $d \equiv 2, 3 \pmod{4}$ or $\mathbb{Z}[\frac{1+\sqrt{d}}{2}]$ if $d \equiv 1 \pmod{4}$.

Given a number field K , a separable extension $L \mid K$, and $\sigma : L \rightarrow \overline{K}$ varying over the K -embeddings of L into the algebraic closure of K , we also have the notion of the norm $\text{Nm}_{L \mid K}$ and trace $\text{Tr}_{L \mid K}$ of an element $\alpha \in L$, given by:

$$\text{Nm}_{L \mid K}(\alpha) := \prod_{\sigma} \sigma(\alpha), \text{ and } \text{Tr}_{L \mid K}(\alpha) := \sum_{\sigma} \sigma(\alpha).$$

A natural question to ask is if $\mathcal{O}_K$ preserves any of the properties of $\mathbb{Z}$, such as being a unique factorisation domain. Indeed, this turns out to be false, as is evident when considering $30 \in \mathbb{Z}[\sqrt{-29}]$. Here, $30 = 2 \cdot 3 \cdot 5 = (1 - \sqrt{-29})(1 + \sqrt{-29})$, where both expressions are products of irreducible elements. A way to remedy this misfortune, due to Dedekind and Kummer, was to note that these rings like $\mathcal{O}_K$ had many of the properties of what would eventually comprise the notion of a Dedekind domain, wherein every nonzero proper ideal factors into primes. The role of ideals was to construct the “ideal objects” to remedy the failure of unique factorisation in extensions.

Theorem 2.1.3. *For $\mathcal{O}$ a Dedekind domain, every proper ideal $(0) \neq \mathfrak{a} \subseteq \mathcal{O}$ can be expressed uniquely as:*

$$\mathfrak{a} = \prod_{i=1}^r \mathfrak{p}_i^{e_i}$$

where $\mathfrak{p}_i \in \text{Spec}(\mathcal{O})$ and $e_i \in \mathbb{Z}_{>0}$.

Another logical question to ask is: given a number field K , how far is $\mathcal{O}_K$ from having unique factorisation? To answer this, we first familiarise ourselves with the notion of a fractional ideal of a Dedekind domain $\mathcal{O}$. Such an ideal $\mathfrak{a}$ is a nonzero $\mathcal{O}$ -submodule of K satisfying:

$$d\mathfrak{a} := \{da : a \in \mathfrak{a}\} \subseteq \mathcal{O}$$

or, in other words, there is a $0 \neq d \in \mathcal{O}$ such that $d\mathfrak{a} \subseteq \mathcal{O}$. For example, $(3/2)\mathbb{Z}$ is a fractional ideal as, for $2 \in \mathbb{Z}$, $(2\mathbb{Z})(3/2\mathbb{Z}) = 3\mathbb{Z} \subseteq \mathbb{Z}$. We denote by $I(\mathcal{O})$ the set of fractional ideals of $\mathcal{O}$, which is a free abelian group on the set of nonzero prime ideals. Then if we denote by $P(\mathcal{O}) \subseteq I(\mathcal{O})$ the subgroup of principal ideals, we define $\text{Cl}(\mathcal{O}) := I(\mathcal{O})/P(\mathcal{O})$ to be the ideal class group of $\mathcal{O}$. Moreover, we write $h_K := |\text{Cl}(\mathcal{O}_K)|$ as the “class number” of K . Exploiting the fact that $\mathcal{O}_K$ has unique factorisation if and only if $\mathcal{O}_K$ is a principal ideal domain, we observe that $h_K = 1$ if and only if $\mathcal{O}_K$ has unique factorisation.

One of the last points to cover is the behaviour of factorisations in extensions of Dedekind domains. Beginning with the example of the Gaussian integers $\mathbb{Z}[i]$, we recall that the behaviour

of primes in this extension is characterised by:

$$\begin{cases} 2\mathbb{Z} = (1+i)^2\mathbb{Z}[i] & \text{i.e. } 2 \text{ ramifies;} \\ p\mathbb{Z} = \mathfrak{p}\bar{\mathfrak{p}}\mathbb{Z}[i] & \text{i.e. } p \text{ splits if } p \equiv 1 \pmod{4}; \\ p\mathbb{Z} = p\mathbb{Z}[i] & \text{i.e. } p \text{ is otherwise inert.} \end{cases}$$

We generalise this notion for finite extensions $L : K$ by noting that $\mathfrak{p} \in \text{Spec}(\mathcal{O}_K)$ decomposes in a unique way into primes $\mathfrak{P}_i \in \text{Spec}(\mathcal{O}_L)$, where $\mathfrak{p}\mathcal{O}_K = \prod_{i=1}^g \mathfrak{P}_i^{e_i} \mathcal{O}_L$. Here, e_i denotes the ramification index of a prime divisor $\mathfrak{P}_i \mid \mathfrak{p}$, and we denote by f_i the inertia degree of $\mathfrak{P}_i$ over $\mathfrak{p}$, given by the degree of the field extension $f_i = [\mathcal{O}_L/\mathfrak{P}_i : \mathcal{O}_K/\mathfrak{p}]$. Indeed, the quotient $\mathcal{O}_K/\mathfrak{p}$ is always a field as prime ideals are maximal in Dedekind domains. As per the example above, a prime which does not split nor ramify is inert. If $L : K$ is separable, and $m = [L : K]$, then these terms are connected via a fundamental identity.

Proposition 2.1.4. *For $L : K$ a separable extension, we have:*

$$\sum_{i=1}^g e_i f_i = m.$$

If $L : K$ is Galois then, for all $1 \leq i \leq g$, $e_i = e$ and $f_i = f$ for some $e, f \in \mathbb{N}$, simplifying the above to $m = efg$.

The introduction of these e_i and f_i terms also allow for a brief introduction to the norm of an ideal and its absolute norm. For a more detailed construction, see [21, p. 68-70]. Here, the idea is to construct a norm map $\underline{\text{Nm}}_{L:K} : I(\mathcal{O}_L) \rightarrow I(\mathcal{O}_K)$ on ideals which behaves in a similar way to the norm of an element $\text{Nm}_{L:K}$. If $\mathfrak{p} \in \text{Spec}(\mathcal{O}_K)$ is principal, $\mathfrak{p} = (p)$, then we ideally want the following to hold

$$\underline{\text{Nm}}((p)\mathcal{O}_L) = (\text{Nm}(p))\mathcal{O}_K = (p^{[L:K]}) = \mathfrak{p}^{[L:K]}.$$

Given that $\underline{\text{Nm}}$ should be a homomorphism, we should also have that

$$\underline{\text{Nm}}(\mathfrak{p}\mathcal{O}_L) = \underline{\text{Nm}}\left(\prod_{i=1}^g \mathfrak{P}_i^{e_i} \mathcal{O}_L\right) = \prod_{i=1}^g \underline{\text{Nm}}(\mathfrak{P}_i)^{e_i} \mathcal{O}_L.$$

Recalling the above proposition leads us to define $\underline{\text{Nm}}$ first on prime ideals as $\underline{\text{Nm}}(\mathfrak{P}_i) := \mathfrak{p}^{f_i}$ where $\mathfrak{P}_i \cap \mathcal{O}_K = \mathfrak{p}$ and extend multiplicatively. On top of this, we have the notion of the numerical norm $\text{Nm} : I(\mathcal{O}_K) \rightarrow \mathbb{Q}_{>0}$ given by $\text{Nm}(\mathfrak{a}) := |\mathcal{O}_K/\mathfrak{a}|$ for $\mathfrak{a} \subseteq \mathcal{O}_K$. If $\mathfrak{a} = (\alpha) \in I(\mathcal{O}_K)$, then

$\text{Nm}(\mathfrak{a}) = |\text{Nm}_{K:\mathbb{Q}}(\alpha)|$ and, by convention, $\text{Nm}((0)) := 0$. We distinguish between the numerical norm and the norm of an element by whether we input ideals or regular elements. Similar to the norm of an element above, for $\mathfrak{P} \subseteq \mathcal{O}_L$ and $\mathfrak{p} = \mathfrak{P} \cap \mathcal{O}_K$, the norm of an ideal also satisfies:

$$\text{Nm}_{L:K}(\mathfrak{P}) = \prod_{\sigma \in \text{Gal}(L|K)} \sigma(\mathfrak{P}) \subseteq \mathcal{O}_K.$$

See [21, p. 69] for a proof of the identity.

2.1.2. Local and Global Fields. By a *global field*, we mean either a finite extension of $\mathbb{Q}$ or the function field of an irreducible algebraic curve over a finite field $\mathbb{F}_p(t)$. On the other hand, a *local field* is a complete, Hausdorff, locally compact topological field with respect to a discrete valuation. This section will intend to briefly explore the interplay between these two concepts. Let K be an algebraic number field. We can equip K with an absolute value $|\cdot| : K \rightarrow \mathbb{R}$ which satisfies:

- a) $|x| > 0$ and $|x| = 0$ if and only if $x = 0$;
- b) $|xy| = |x||y|$;
- c) $|x + y| \leq |x| + |y|$.

We note that, if we replace the third case with $|x + y| \leq \max\{|x|, |y|\}$, then we get what is known as a *nonarchimedean absolute value*.

On top of the standard absolute value, which we will denote as $|\cdot|_\infty$, we also define the p -adic absolute value on $\mathbb{Q}$ as $|\alpha|_p := p^{-v_p(\alpha)}$ where, if $\alpha = p^n \frac{a}{b}$, with $p \nmid a, b$, then $v_p(\alpha) = n$. Here, $v_p : \mathbb{Q} \rightarrow \mathbb{Z}_{\geq 0} \cup \{\infty\}$ is called the p -adic valuation. Just as $\mathbb{Q}$ can be completed as $\mathbb{R}$ with respect to $|\cdot|_\infty$, we can define a completion of $\mathbb{Q}$ with respect to $|\cdot|_p$, denoted by $\mathbb{Q}_p$, as the set of equivalence classes of Cauchy sequences of rational numbers. Such a completion then inherits the topology induced by the metric $d_p(x, y) := |x - y|_p$, essentially equating the closeness of $x, y \in \mathbb{Q}_p$ with how “divisible” by p the number $x - y$ is.

Example 2.1.5. An example of an element of $\mathbb{Q}_p$, say for $p = 5$, is the series $(\dots 444)_5 = \sum_{i \geq 0} 4 \cdot 5^i$, which converges in $\mathbb{Q}_5$ as

$$\left| \sum_{i \geq 0} 4 \cdot 5^i - \sum_{n \geq i \geq 0} 4 \cdot 5^i \right|_5 = |4 \cdot 5^{n+1} + 4 \cdot 5^{n+2} + \dots|_5 = (1/5)^{n+1}$$

which tends to 0 as $n \rightarrow \infty$. Given that $\mathbb{Q}_5$ is complete, $\sum_{i \geq 0} 4 \cdot 5^i \in \mathbb{Q}_5$ and, in fact, it converges to -1 . This reflects the notion that the successive approximations of the sum above, $S_n = \sum_{i \geq 0} 4 \cdot 5^i$, have the property that they are all congruent to $-1 \pmod{5^n}$.

We can also define the ring $\mathbb{Z}_p$, the p -adic integers, to be the elements $\alpha \in \mathbb{Q}_p$ which satisfy $|\alpha|_p \leq 1$. Here, $\mathbb{Z}_p$ is a compact neighbourhood of $\mathbb{Q}_p$, being a closed unit ball around 0 of the locally compact $\mathbb{Q}_p$. Whilst fascinating in its own right, we will refrain from exploring the topology and arithmetic of $\mathbb{Q}_p$ and $\mathbb{Z}_p$ within this chapter, and suggest that the reader considers [27, p. 175-177] and [15, p. 1-20] for a further exploration.

We now briefly introduce the projective limit by building off Example 2.1.5 in a similar vein to [2, p. 298-301]. The above example indicates that there exists a solution to $x^2 = -1$ in $\mathbb{Z}_5 \subset \mathbb{Q}_5$. We will construct this solution by the means of the projective limit. Over $\mathbb{Z}$, whilst $x^2 = -1$ has no solutions, there are two such solutions modulo 5 (namely $x_1 \equiv 2, 3 \pmod{5}$). Taking the first solution, as they are the same up to a sign, and noting that any such $x_1 \in \mathbb{Z}$ is of the form $x_1 = 5x_2 + 2$. Substituting this into our desired equation then yields $25x_2^2 + 20x_2 = -5$ which, when considered modulo 25 yields $20x_2 \equiv -5 \pmod{25}$, or $4x_2 \equiv -1 \pmod{5}$, where $x_2 \equiv 1 \pmod{5}$. Then x_1 modulo 25 is given by $x_1 = 5x_2 + 2 \equiv 5 + 2 \pmod{25} \equiv 7 \pmod{25}$.

Consider the sequence of ring homomorphisms $f_i : \mathbb{Z}/5^{i+1}\mathbb{Z} \rightarrow \mathbb{Z}/5^i\mathbb{Z}$:

$$\dots \xrightarrow{f_{i+1}} \mathbb{Z}/5^{i+1}\mathbb{Z} \xrightarrow{f_i} \mathbb{Z}/5^i\mathbb{Z} \xrightarrow{f_{i-1}} \dots \xrightarrow{f_1} \mathbb{Z}/5\mathbb{Z},$$

which satisfy $f(x_{i+1}) = x_i$. These homomorphisms then allow us to repeat this process indefinitely by lifting each x_i to x_{i+1} by our prior calculations, and the homomorphisms f_i ensure coherency within the sequence $(x_n)_{n \geq 1}$. We write

$$\varprojlim \mathbb{Z}/5^i\mathbb{Z} := \{\underline{x} = (\dots, x_2, x_1) \in \prod_i \mathbb{Z}/5^i\mathbb{Z} \mid f_i(x_{i+1}) = x_i\}$$

to be the set of such sequences, which forms a ring under component-wise addition and multiplication. In fact, the above is isomorphic to the 5-adic integers $\mathbb{Z}_5$. We consolidate the above with a formal definition of the projective limit.

Definition 2.1.6. Let X be a topological space with subsets $\{X_i\}_{i \in I}$ with $X_i \subseteq X_j$ for $i \leq j$. Moreover, let $\{f_{ij}\}_{i,j \in I}$ be a family of continuous maps $f_{ij} : X_j \rightarrow X_i$ such that $f_{ii} = \text{id}_{X_i}$ and

$f_{ik} = f_{ij} \circ f_{jk}$. Then the projective limit $\tilde{X} := \varprojlim X_i$ is defined to be the subset

$$\tilde{X} := \left\{ (x_i)_{i \in I} \in \prod_i X_i \mid f_{ij}(x_j) = x_i \text{ for } i \leq j \right\} \subseteq \prod_i X_i.$$

By a “place” ν of K , denoted $\nu \in \text{Pl}(K)$, we refer to an equivalence of absolute values. More explicitly, $|\cdot|_1$ and $|\cdot|_2$ are equivalent if they induce the same topology on K or if $|\cdot|_2 = |\cdot|_1^a$ for some $a > 0$. Then we say that these two absolute values are of the same place $\nu \in \text{Pl}(K)$. This leads us to an important theorem on the classification of places of a number field.

Theorem 2.1.7. *For K a number field, there is exactly one place $\nu \in \text{Pl}(K)$ for each:*

- a) non-zero prime ideal $\mathfrak{p} \in \text{Spec}(\mathcal{O}_K)$;
- b) real and (conjugate pair of) complex embeddings $\sigma, \bar{\sigma} : K \hookrightarrow \mathbb{R}, \mathbb{C}$.

Proof. See [21, p. 111] □

As a result, for each finite place $\nu = \mathfrak{p} \in \text{Spec}(\mathcal{O}_K)$, $\mathfrak{p} \neq (0)$, we set $|\alpha|_{\mathfrak{p}} := (1/\text{Nm}(\mathfrak{p}))^{\nu_{\mathfrak{p}}(\alpha)}$. When $\nu = \infty$, we have $|\alpha|_{\sigma} = |\sigma(\alpha)|_{\infty}$ for each real embedding, and $|\alpha|_{\sigma} = |\sigma(\alpha)|^2$ for each (non-real) complex embedding. We denote by K_{ν} the completion of K with respect to $|\cdot|_{\nu}$, and $\mathcal{O}_{K_{\nu}}$ its ring of integers in K_{ν} . Whilst we could consider any other representative from each place $\nu \in \text{Pl}(K)$, these normalised values are considered to be the most natural definitions for the following “product formula” to hold.

Theorem 2.1.8. *For each place $\nu \in \text{Pl}(K)$, and normalised absolute values $|\cdot|_{\nu}$:*

$$\prod_{\nu} |\alpha|_{\nu} = 1$$

for every $\alpha \in K^{\times}$.

Proof. See [21, p. 138]. □

We conclude this chapter with a quick introduction to the notion of a uniformising parameter and the concept of the projective limit. For a discrete nonarchimedean absolute value field K , we choose $\pi \in K$ such that $|\pi|$ the largest value in the set $\{|\alpha| : \alpha \in K, |\alpha| < 1\}$. This generates the unique maximal ideal in the valuation ring $R := \{\alpha \in K : |\alpha| \leq 1\}$.

After completing K at some finite place $\nu \in \text{Pl}(K)$, the absolute value on K can be extended to one on K_{ν} , maintaining the property of being a discrete absolute value. Completion also preserves

the chosen uniformiser π , where the completed maximal ideal remains (π) . The uniformiser π may be useful in computing the valuation of any element in $x \in K$, which can be expressed uniquely as $x = u\pi^n$, and characterising the ideals of A_K with the set $\{(\pi)^n : n \in \mathbb{Z}\}$.

2.2. Ideas from the Étale Cohomology and the Motivic World

We conclude the preliminaries with a brief motivation for motives, a concept introduced by Alexandre Grothendieck in the 1960s to unify the vast yet similarly behaved theories of cohomology. Alongside the notion of a motive, we will also briefly introduce the notion of the étale cohomology. Whilst both concepts are fascinating in their own right, we will refrain from an extensive exploration of these topics as they are beyond the scope of this paper. Rather, we will intend to “motivate” motives and related topics in order to introduce Jacobi and Jacket motives later on in the text. For a more extensive exploration of these topics, one may consider [26], [20] and [19].

Given an algebraic variety X over $\mathbb{C}$, or the set of zeros to a collection of polynomial equations, its set of points admits a complex manifold along with the additional structure of the automorphism of complex conjugation, meaning that the vector spaces associated with the singular cohomology groups $H^i(X(\mathbb{C}))$ inherit the action of $\text{Gal}(\mathbb{C}|\mathbb{R})$. In this spirit, the étale cohomology was introduced so that, for a variety X defined over $\mathbb{Q}$, the cohomology groups $H_{\text{ét}}^i(X)$ carry a natural action of the absolute Galois group $\text{Gal}(\overline{\mathbb{Q}} : \mathbb{Q})$.

Let X be a variety over K . In Serre’s paper [26] he explores Grothendieck’s motivation for motives. The idea is to first construct a $\mathbb{Q}$ -abelian category (or, equivalently, one which satisfies the snake lemma), denoted $\underline{\mathcal{M}}(K)$. Along with this, we have a contravariant functor h whose output decomposes into a direct sum of objects in $\underline{\mathcal{M}}(K)$: $X \mapsto h(X) = \bigoplus h^i(X)$. Moreover, $\underline{\mathcal{M}}(K)$ must satisfy the following properties:

- a) Given the objects A, B of $\underline{\mathcal{M}}(K)$, $\text{Hom}_{\mathbb{Q}}(A, B)$ is a finite dimensional $\mathbb{Q}$ -vector space.
- b) For each $\ell \neq \text{char}(K)$, there exists a functor $T_{\ell} : \underline{\mathcal{M}}(K) \rightarrow \text{Gal}(\overline{K}|K)\text{-}\mathbb{Q}_{\ell}$ representations such that $T_{\ell}(h^i(X)) = H_{\text{ét}}^i(\overline{X}, \mathbb{Q}_{\ell})$, where $\overline{X}$ denotes the variety X over the algebraic closure of K .

If $\text{char}(K) = 0$ or if K embeds into $\mathbb{C}$, then the second property may be replaced with the de Rham or Betti cohomology respectively [26, p. 334-335].

The main reason for exploring motives will be evident when considering Jacobi motives $\mathbf{J}(\theta)$, as per their characterisation in [30, p. 4, 11-12]. Whilst ultimately a notational device in this thesis, the motivic framework remains valuable for organising the relationships between algebraic objects and the arithmetic information they encode.

CHAPTER 3

Jacobi Sums and Grossencharacters

The purpose of this section is to establish the fundamental connection between Jacobi sums and Hecke Grossencharacters (also referred to as Hecke characters or merely Grossencharacters) defined in this case over cyclotomic fields. Around 1837, the mathematician Peter Gustav Dirichlet introduced the concept of Dirichlet characters in order to prove his theorem on prime numbers in arithmetic progression by studying the behaviour of the character's associated L -function [12, p. 144-150]. Then in the early 1900's, Erich Hecke introduced the notion of a Grossencharacter to further Dirichlet's exploration of L -functions in the setting of more general number fields. This section expands on André Weil's 1952 paper on Jacobi sums [31], coming from point counts on polynomial equations over finite fields, and Hecke characters.

3.1. Definitions

We begin by first introducing the necessary definitions and relevant results pertaining to the study of Jacobi sums and Grossencharacters. Then we will explore a reformulation of Weil's original proof that Jacobi sums are indeed characters in the sense of Hecke. Throughout this section, we will let K be a number field which, unless otherwise specified, will generally be the cyclotomic extension $\mathbb{Q}(\zeta_m)$, and $\mathcal{O}_K = \mathbb{Z}[\zeta_m]$ its ring of integers. For an ideal $(0) \neq \mathfrak{a} \subseteq \mathcal{O}_K$, we denote $I_K(\mathfrak{a})$ as the set of ideals in $\mathcal{O}_K$ prime to $\mathfrak{a}$, and we use the unadorned $I(\mathfrak{a})$ when the field K is clear.

Definition 3.1.1. *Let G be a group. A (unitary) character of a group is a homomorphism $\chi : G \rightarrow \mathbb{C}^\times$. Depending on whether G is written multiplicatively or additively, we say that χ is a multiplicative or additive character respectively.*

If $G = \mathbb{F}_q^\times$, where q is some power of a prime number p , the multiplicative character χ is extended to the whole field $\mathbb{F}_q$ by setting $\chi(0) = 0$.

The notion of a Dirichlet character will also be of use when later studying Hecke's Grossencharacters.

Definition 3.1.2. We say that $\chi : \mathbb{Z} \rightarrow \mathbb{C}^\times$ is a Dirichlet character (modulo m) if there exists $m \in \mathbb{Z}_{\geq 1}$ such that χ satisfies the following properties:

- a) $\chi(n + m) = \chi(n)$ for all $n \in \mathbb{Z}$;
- b) $\chi(kn) = \chi(k)\chi(n)$ for all $k, n \in \mathbb{Z}$;
- c) $\chi(n) \neq 0$ if and only if $\gcd(m, n) = 1$.

In addition to the notion of a multiplicative and Dirichlet character, we will also introduce the concept of the Hecke Grossencharacter in a similar vein to [9, p. 35]. Along with the set $I(\mathfrak{f})$ defined above, we consider $P(\mathfrak{f})$, which consists of the subgroup of the principal ideals coprime to $\mathfrak{f} \subseteq \mathcal{O}_K$. Then we say that the ray class group is given by the quotient $\text{Cl}(\mathfrak{f}) := I(\mathfrak{f})/P(\mathfrak{f})$. This allows us to first define Hecke Grossencharacters on principal ideals and then to extend it to all ideals $\mathfrak{a} \in I(\mathfrak{f})$.

Definition 3.1.3. Given a number field K , where $n = [K : \mathbb{Q}] = r + 2s$ and $r, 2s$ are the number of unique real and complex embeddings of $K \hookrightarrow \mathbb{R}, \mathbb{C}$, respectively. With $I(\mathfrak{f})$ given as before, a Hecke Grossencharacter $\psi : I(\mathfrak{f}) \rightarrow \mathbb{C}^\times$ of K is a complex-valued function which satisfies the following two conditions:

- (i) it is multiplicative for all $\mathfrak{a}, \mathfrak{b} \in I(\mathfrak{f})$;
- (ii) for all $\alpha \in \mathcal{O}_K$ and $\alpha \equiv 1 \pmod{\mathfrak{f}}$, there exist rational integers $\{e_i\}_{1 \leq i \leq r+s}$ and rational numbers $\{c_i\}_{1 \leq i \leq r+s}$ such that $\psi((\alpha))$ decomposes as:

$$(3.1.4) \quad \psi((\alpha)) = \prod_{i=1}^{r+s} \alpha^{e_i \sigma_i} |\alpha|^{c_i \sigma_i},$$

where the collection $\{\sigma_i\}_{i=1, \dots, r+s}$ denotes the non-equivalent complex embeddings $K \xrightarrow{\sigma_i} \mathbb{C}$, and $(\cdot)^{\sigma_i}$ denotes the right action of σ_i on α .

Example 3.1.5. We construct a Grossencharacter $\mathcal{G}$ on $\mathbb{Q}(i)$ in the following way. Given $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$ is a PID, we merely have to define where $\mathcal{G}$ sends prime ideals of the form $(a + bi)$. To ensure the uniqueness of the representation of $(a + bi)$, we enforce that $a \equiv 1 \pmod{2}$, and set $\mathcal{G}((a + bi)) = a + bi$ and extend multiplicatively to obtain the Grossencharacter $\mathcal{G}$ on $I_{\mathbb{Q}(i)}((2))$.

Example 3.1.6. Now, we consider the case where $\mathcal{O}_K$ is not a PID. Let $K = \mathbb{Q}(\sqrt{-5})$, whose class group is nontrivial: $\text{Cl}(K) \cong (\mathbb{Z}/2\mathbb{Z})^\times$. We set $\psi_\infty(\mathfrak{a}) = \text{Nm}(\mathfrak{a})$ to be the infinity type of this Grossencharacter. Then for $\mathfrak{g}$ a generator of $\text{Cl}(K)$, we may set $\tilde{\psi}(\mathfrak{g}) = -1$, where $\tilde{\psi}$ was given above, allowing us to extend ψ to all ideals in $I((3))$ by $\psi(\mathfrak{a}) = \tilde{\psi}([\mathfrak{a}]) \text{Nm}(\mathfrak{a})$.

Before exploring the formalisation of the Jacobi sum, we first recall the setup from Weil's paper [31, p. 487-489]. Let m be a positive integer and ζ_m an m^{th} root of unity. We consider the cyclotomic extension $\mathbb{Q}(\zeta_m) \mid \mathbb{Q}$ and its corresponding Galois group $\text{Gal}(\mathbb{Q}(\zeta_m) \mid \mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$. Take a prime ideal $\mathfrak{p} \in I((m^2))$, and set $q = \text{Nm}(\mathfrak{p})$, where $q \equiv 1 \pmod{m}$. A residue field of $\mathcal{O}_{\mathbb{Q}(\zeta_m)} = \mathbb{Z}[\zeta_m]$ is then constructed via defining $\mathbb{F}_{\mathfrak{p}}$ to be the quotient $\mathbb{F}_{\mathfrak{p}} := \mathbb{Z}[\zeta_m]/\mathfrak{p} \cong \mathbb{F}_q$ for q a power of a prime, which is indeed a field as prime ideals are maximal in Dedekind domains.

We define the multiplicative character $\chi_{\mathfrak{p}} : \mathbb{F}_{\mathfrak{p}} \rightarrow \mathbb{C}^1$ as:

$$(3.1.7) \quad \chi_{\mathfrak{p},m}(x) = \zeta_m^r$$

where $r \in \mathbb{Z}$ ultimately depends on the variable x and satisfies $\zeta_m^r \equiv x^{(q-1)/m} \pmod{\mathfrak{p}}$. Then $\mathfrak{p}, m$ is extended to all elements of $\mathbb{F}_{\mathfrak{p}}$ when necessary by setting $\chi_{\mathfrak{p}}(0) := 0$. Moreover, for $\sigma \in \text{Gal}(\mathbb{Q}(\zeta_m) \mid \mathbb{Q})$ the multiplicative character in 3.1.7 has the following property:

$$(3.1.8) \quad \chi_{\sigma(\mathfrak{p}),m}(\sigma(x)) \equiv \sigma(x)^{(q-1)/m} \pmod{\sigma(\mathfrak{p})} \equiv \zeta_m^{kr} = \sigma(\zeta_m^r) = \sigma(\chi_{\mathfrak{p},m}(x)) \pmod{\mathfrak{p}}$$

due to the fact that $\chi_{\mathfrak{p},m}(x) - \zeta_m^r \in \mathfrak{p}$ yields $\sigma(\chi_{\mathfrak{p}}(x)) - \zeta_m^{kr} \in \sigma(\mathfrak{p})$ and the canonical isomorphism:

$$(3.1.9) \quad \Psi_\sigma : \mathbb{F}_{\mathfrak{p}} \rightarrow \mathbb{F}_{\sigma(\mathfrak{p})}, \quad \alpha \mapsto \sigma(\alpha),$$

as automorphisms of the Galois group act transitively on the prime factors of an ideal.

We finalise this section with a quick discussion of the trivial character. When necessary, we may make use of two different conventions of trivial characters, denoting the usual one by $\varepsilon : \mathbb{F}_{\mathfrak{p}} \rightarrow \mathbb{C}^1$ and the ‘‘Ireland and Rosen’’ trivial character by $\eta : \mathbb{F}_{\mathfrak{p}} \rightarrow \mathbb{C}^1$. Whilst $\varepsilon(x) = \eta(x) = 1$ for all $x \in \mathbb{F}_{\mathfrak{p}}^\times$, we extend $\varepsilon(0) := 0$ in the usual way (see: [9, p. 13] and [14, p. 10]) and let $\eta(0) := 1$ as per [13, p. 88].

Remark 3.1.10. Despite ε being the more standard choice of trivial character in the more contemporary literature, the presence of the Ireland and Rosen trivial character η will allow us to translate the theorems in [13] pertaining to point counts with greater ease for use in this thesis.

3.2. Gauss and Jacobi Sums

Before introducing the Gauss sum, we consider the additive character $\Theta_0 : \mathbb{F}_p \rightarrow \mathbb{C}^1$ given by $\Theta_0(x) = e^{2\pi i x/p}$. Without loss of generality, $p \in \mathbb{Z}_{>0}$ is chosen to be a prime number such that $p \nmid m$. One may then extend Θ_0 from $\mathbb{F}_p$ to all of $\mathbb{F}_{\mathfrak{p}}$ via the composition $\Theta := \Theta_0 \circ \text{Tr}_{\mathbb{F}_{\mathfrak{p}} | \mathbb{F}_p} : \mathbb{F}_{\mathfrak{p}} \rightarrow \mathbb{C}^1$ given $\text{Tr}_{\mathbb{F}_{\mathfrak{p}} | \mathbb{F}_p}(x) \in \mathbb{F}_p$ for all $x \in \mathbb{F}_{\mathfrak{p}}$.

Definition 3.2.1. Let $\Theta : \mathbb{F}_{\mathfrak{p}} \rightarrow \mathbb{C}^1$ be the additive character defined above. For $a \in (\mathbb{Z}/m\mathbb{Z})^\times$, we define the Gauss sum as:

$$(3.2.2) \quad g_{a/m}(\mathfrak{p}) = \sum_{x \in \mathbb{F}_{\mathfrak{p}}} \chi_{\mathfrak{p},m}(x)^a \Theta(x).$$

We accompany Definition 3.2.1 with a lemma exploring how the Gauss sum behaves when one sets $\chi = \varepsilon$ or $\chi = \eta$.

Lemma 3.2.3. Let $g(\varepsilon)$ and $g(\eta)$ denote the Gauss sum of the trivial character ε and the “Ireland and Rosen” trivial character η respectively. Then $g(\varepsilon) = -1$ and $g(\eta) = 0$.

Proof. We begin with the summation of $g(\eta) = \sum_x \eta(x) \Theta(x) = \sum_x \Theta(x)$. We fix $y \in \mathbb{F}_{\mathfrak{p}}$ such that $y \notin \ker(\Theta)$, which must exist due to the construction of Θ as a non-trivial additive character. We observe that $\Theta(y)g(\eta) = \sum_x \Theta(y) \Theta(x) = \sum_x \Theta(x+y) = \sum_t \Theta(t) = g(\eta)$ where a summation change is made by letting $t = x+y$. Hence, $g(\eta)(\Theta(y) - 1) = 0$ and, given $y \notin \ker(\Theta)$, this forces $g(\eta) = 0$.

To finalise the proof, we note that $g(\varepsilon) + 1 = \sum_{x \in \mathbb{F}_{\mathfrak{p}}^\times} \Theta(x) + 1 = \sum_{x \in \mathbb{F}_{\mathfrak{p}}} \Theta(x) = 0$ due to the previous observation. Hence, $g(\varepsilon) = -1$. $\square$

We now introduce the notion of the Jacobi sum, closely resembling that of Weil’s paper, before moving on to a more general construction later on.

Definition 3.2.4. Let $\chi_{\mathfrak{p}}$ be the character from 3.1.7 and $\underline{a} \in G^r$. We call the function $J_{\underline{a}} : I((m^2)) \rightarrow \mathbb{C}$ given by:

$$(3.2.5) \quad J_{\underline{a}}(\mathfrak{p}) = (-1)^{r+1} \sum_{x_1, \dots, x_r} \prod_{i=1}^r \chi_{\mathfrak{p}}(x_i)^{a_i}$$

a Jacobi sum, where the summation is taken over the hypersurface $1 + \sum_{i=1}^r x_i = 0$ for $(x_1, \dots, x_r) \in \mathbb{A}^r(\mathbb{F}_{\mathfrak{p}})$. The Jacobi sum is then extended from prime ideals $\mathfrak{p} \in I((m^2))$ to all ideals in $I((m^2))$

via the relation

$$(3.2.6) \quad J_{\underline{a}}(\mathfrak{a}\mathfrak{b}) = J_{\underline{a}}(\mathfrak{a})J_{\underline{a}}(\mathfrak{b}).$$

Remark 3.2.7. In the next section when we discuss the reformulation of Jacobi sum in terms of Gauss sums, we will assume the fact that it is invariant of the choice of additive character. This fact will be proven in the next chapter (see Theorem 4.2.1).

3.3. Reframing Jacobi Sums

As was previously mentioned, the sum given in 3.2.5 is the same notion of the Jacobi sum which Weil utilised in his original proof that Jacobi sums are Grossencharacters. In order to translate Weil's original exploration of the Jacobi sum into the more contemporary language of Anderson's formalism [1], we will first demonstrate that the sum $J_{\underline{a}}$ can be expressed as a product of Gauss sums. We start by considering a lemma to help us in the proof of the proposition below.

Lemma 3.3.1. *Let $\Theta : \mathbb{F}_{\mathfrak{p}} \rightarrow \mathbb{C}^1$ be the non-trivial additive character as per 3.2.1. Then*

$$(3.3.2) \quad \left(\frac{1}{q}\right) \sum_{\lambda \in \mathbb{F}_{\mathfrak{p}}} \Theta(\lambda(x - y)) = \delta(x, y),$$

where the $\delta(x, y)$ on the right-hand side is a Kronecker delta symbol.

Proof. See [13, p. 143]. □

Proposition 3.3.3. *Let $s = -\sum_{i=1}^r a_i$. For the Jacobi sum $J_{\underline{a}}(\mathfrak{p})$ and the Gauss sum $g_{a/m}(\mathfrak{p})$ from 3.2.5 and 3.2.1 respectively, we have that:*

$$(3.3.4) \quad J_{\underline{a}}(\mathfrak{p}) = \frac{(-1)^{r+1}}{q} g_{s/m}(\mathfrak{p}) \prod_{i=1}^r g_{a_i/m}(\mathfrak{p}).$$

Proof. Let $x = \sum_{i=1}^r x_i$ and $y = -1$. Plugging both x and y into the above identity (Lemma 3.3.1), multiplying both sides by $(-1)^{r+1} \prod_{i=1}^r \chi_{\mathfrak{p}}(x_i)^{a_i}$, and summing both sides over $(x_1, \dots, x_r) \in$

$\mathbb{A}^r(\mathbb{F}_p)$, we have:

$$\begin{aligned}
J_{\underline{a}}(\mathfrak{p}) &= (-1)^{r+1} \sum_{(x_1, \dots, x_r) \in \mathbb{A}^r(\mathbb{F}_p)} \prod_{i=1}^r \chi_{\mathfrak{p}}(x_i)^{a_i} \delta \left(\sum_{j=1}^r x_j, -1 \right) \\
&= \frac{(-1)^{r+1}}{q} \sum_{(x_1, \dots, x_r) \in \mathbb{A}^r(\mathbb{F}_p)} \prod_{i=1}^r \chi_{\mathfrak{p}}(x_i)^{a_i} \sum_{\lambda \in \mathbb{F}_p} \Theta(\lambda) \prod_{j=1}^r \Theta(\lambda x_j) \\
&= \frac{(-1)^{r+1}}{q} \sum_{\lambda \in \mathbb{F}_p} \Theta(\lambda) \sum_{x_1, \dots, x_r} \prod_i \chi_{\mathfrak{p}}(x_i)^{a_i} \Theta(\lambda x_i),
\end{aligned}$$

where we first focus on the quantity $\sum_{x_1, \dots, x_r} \prod_i \chi_{\mathfrak{p}}(x_i)^{a_i} \Theta(\lambda x_i) = \prod_i \sum_{x_i} \chi_{\mathfrak{p}}(x_i)^{a_i} \Theta(\lambda x_i)$ greatly resembling a Gauss sum. One may observe that, via a summation change from $(x_i)_{1 \leq i \leq r} \in \mathbb{A}^r(\mathbb{F}_p)$ to $(t_i)_{1 \leq i \leq r} \in \mathbb{A}^r(\mathbb{F}_p)$ where $t_i = \lambda x_i$ for $\lambda \neq 0$, the following holds:

$$\begin{aligned}
\prod_i \sum_{x_i} \chi_{\mathfrak{p}}(x_i)^{a_i} \Theta(\lambda x_i) &= \prod_i \sum_{t_i} \chi_{\mathfrak{p}}(\lambda^{-1} t_i)^{a_i} \Theta(t_i) \\
&= \prod_i \sum_{t_i} \chi_{\mathfrak{p}}(\lambda^{-1})^{a_i} \chi_{\mathfrak{p}}(t_i)^{a_i} \Theta(t_i) \\
&= \chi_{\mathfrak{p}}(\lambda)^s \prod_i \sum_{t_i} \chi_{\mathfrak{p}}(t_i)^{a_i} \Theta(t_i) = \chi_{\mathfrak{p}}(\lambda)^s \prod_i g_{a_i/m}(\mathfrak{p}).
\end{aligned}$$

If $\lambda = 0$, then $\Theta(\lambda x_i) = 0$, meaning that we can safely ignore this case.

Finally, we substitute this observation back into the Jacobi sum to conclude that

$$\begin{aligned}
J_{\underline{a}}(\mathfrak{p}) &= \frac{(-1)^{r+1}}{q} \prod_i g_{a_i/m}(\mathfrak{p}) \sum_{\lambda \in \mathbb{F}_p} \chi_{\mathfrak{p}}(\lambda)^s \Theta(\lambda) \\
&= \frac{(-1)^{r+1}}{q} g_{s/m}(\mathfrak{p}) \prod_i g_{a_i/m}(\mathfrak{p}).
\end{aligned}$$

□

Having established that $J_{\underline{a}}(\mathfrak{p}) = ((-1)^{r+1}/q) g_{s/m}(\mathfrak{p}) \prod_{i=1}^r g_{a_i/m}(\mathfrak{p})$, we can now reformulate Jacobi sums in the more contemporary language of Anderson's formalism, also reflecting its modern implementation in Magma (see: [25, p. 18-20] and [30, p. 3-4] for similar constructions). We start by considering formal integral sums $\theta \in \mathbb{Z}[\mathbb{Q}/\mathbb{Z}]$ whose projected sum to $\mathbb{Q}/\mathbb{Z}$ is zero [30, p. 3]. We write $\theta = \sum_i n_i \{y_i\}$, where $n_i \in \mathbb{Z}$ and the choice of the representative of $y_i \in \mathbb{Q}/\mathbb{Z}$ is to be merely its fractional part $\{y_i\} \in \mathbb{Q} \cap [0, 1)$. Writing each $\{y_i\} = y'_i/m_i$ in lowest terms, for

$y'_i, m_i \in \mathbb{Z}$ with $m_i > 0$, we set $m = \text{lcm}(m_1, \dots, m_r)$. We then write

$$(3.3.5) \quad J_\theta(\mathfrak{p}) = \prod_{i=1}^r g_{y_i}(\mathfrak{p})^{n_i}.$$

We consolidate the previous observation with a definition of the *formal Jacobi sum*.

Definition 3.3.6. Let $\theta = \sum_{i=1}^r n_i \{y_i\}$ be the formal sum in $\mathbb{Z}[\mathbb{Q}/\mathbb{Z}]$ whose sum is projected to zero. The formal Jacobi sum is a map $J_\theta : I((m^2)) \rightarrow \mathbb{C}$ where, for prime ideals $\mathfrak{p} \in I((m^2))$:

$$(3.3.7) \quad J_\theta(\mathfrak{p}) = \prod_{i=1}^r g_{y_i}(\mathfrak{p})^{n_i}$$

We extend J_θ multiplicatively to all ideals in $I((m^2))$ by $J_\theta(\mathfrak{a}\mathfrak{b}) = J_\theta(\mathfrak{a})J_\theta(\mathfrak{b})$, for $\mathfrak{a}, \mathfrak{b} \in I((m^2))$.

Remark 3.3.8. For prime powers of ideals $\mathfrak{p}^r \in I((m^2))$, $r \in \mathbb{Z}_{>0}$, we set

$$J_\theta(\mathfrak{p}^r) = J_\theta(\mathfrak{p})^r = \prod_{i=1}^r g_{y_i}(\mathfrak{p})^{rn_i} = J_{r\theta}(\mathfrak{p}).$$

Then for $\mathfrak{a} \in I(m)$ whose decomposition is given by $\mathfrak{a} = \prod_{i=1}^z \mathfrak{p}_i^{e_i}$ for distinct prime ideals $\mathfrak{p}_i$ and natural numbers e_i , we have:

$$J_\theta(\mathfrak{a}) = \prod_{i=1}^z J_{e_i\theta}(\mathfrak{p}_i).$$

Remark 3.3.9. Whilst $\theta \in \mathbb{Z}[\mathbb{Q}/\mathbb{Z}]$ may be given by the sum $\theta = \sum_i n_i \{y_i\}$, we may also consider it as a multiset of defining data of J_θ . To avoid confusion between the two, we will often denote this defining data by S . For example, if $\theta = -3\{1/3\} + \{1/5\} + \{4/5\}$, then

$$S = \{-1/3, -1/3, -1/3, 1/5, 4/5\}.$$

We now introduce a greater level of formalism by introducing the Jacobi motive (see section 2.2 for a brief introduction to motives). In [25, p. 48], Schappacher notes that any algebraic Hecke character (meaning that its values lay in an algebraic number field) ψ comes from a motive $h(\psi)$. Whilst we are yet to prove that these formal Jacobi sums are indeed Hecke Grossencharacters, we introduce the Jacobi motive $\mathbf{J}(\theta)$ to be the motive corresponding to the function $J_\theta(\mathfrak{p})$. Here, addition of these formal sums $\theta_1, \theta_2 \in \mathbb{Z}[\mathbb{Q}/\mathbb{Z}]$ corresponds to the following:

$$\mathbf{J}(\theta_1 + \theta_2) = \mathbf{J}(\theta_1) \otimes_K \mathbf{J}(\theta_2),$$

where K is the field for which both $\mathbf{J}(\theta_1)$ and $\mathbf{J}(\theta_2)$ are defined in. A reason for introducing this framework is to introduce the notion of a Jacket motive.

Definition 3.3.10. We define a Jacket motive $\mathbf{J}(\theta, t, \rho, m)$ to be the Jacobi motive $\mathbf{J}(\theta)$ twisted by the Kummer twist of t^ρ and the m^{th} -Tate twist. More explicitly, we denote a the Kummer-twisted Jacobi motive by t^ρ to be realised as the Jacobi sum

$$J_\theta^{t^\rho}(\mathfrak{p}) := \chi_\mathfrak{p}(t^\rho) J_\theta(\mathfrak{p}),$$

and the m^{th} -Tate twist $J_\theta(\mathfrak{p})(m)$ to be realised as

$$J_\theta(m)(\mathfrak{p}) := \frac{1}{\text{Nm}(\mathfrak{p})^m} J_\theta(\mathfrak{p}).$$

Remark 3.3.11. Throughout the rest of the paper, we will more often than not realise these Jacket motives as “twisted” Jacobi sums with the notation above.

Assuming true for now that J_θ is a Grossencharacter, given that the numerical ideal norm is a Hecke character, it is immediate that a Tate-twisted Jacobi sum should also be a Hecke character. We claim that any Kummer-twisted Jacobi sum should also be a Grossencharacter by arguing that the assignment $\mathfrak{p} \mapsto \chi_\mathfrak{p}(c)$ is a Hecke character for c a fixed element of $\mathbb{F}_\mathfrak{p}^\times$.

Theorem 3.3.12. For $\mathfrak{p}$ a prime ideal of $\mathbb{Z}[\zeta_m]$, the assignment $\mathfrak{p} \mapsto \chi_\mathfrak{p}(c)$ is a Hecke Grossencharacter for $c \in \mathbb{F}_\mathfrak{p}^\times$.

Proof. Given $K = \mathbb{Q}(\zeta_m)$, set $L = K(\sqrt[m]{c})$. Kummer theory provides the framework for us to deduce that, for any such $\eta = \sqrt[m]{c}$, there is a homomorphism from $\text{Gal}(L|K)$ to μ_m given by $\sigma \mapsto \sigma(\eta)/\eta$.

Due to $L | K$ being an abelian extension, for any $\mathfrak{P}_i$ in the factorisation $\mathfrak{p}\mathcal{O}_K = \prod_{i=1}^g \mathfrak{P}_i^{e_i}$, $\text{Frob}_{\mathfrak{P}_i} \equiv \text{Frob}_\mathfrak{p}$ as automorphisms of $\text{Gal}(L|K)$. The Frobenius map acts trivially on K and permutes the roots of $x^m - c$, meaning that $\text{Frob}_\mathfrak{p}(\sqrt[m]{c})/\sqrt[m]{c} = \zeta_m^{s(\mathfrak{p})}$ for some $s(\mathfrak{p}) \in \mathbb{Z}$ depending on $\mathfrak{p}$.

By construction, this is precisely $\chi_\mathfrak{p}(c)$, as if $\text{Nm}(\mathfrak{p}) = q$, then $\eta^{mq} \equiv c \pmod{\mathfrak{p}}$, which yields $\eta^{q-1} \equiv \zeta_m^r \pmod{\mathfrak{p}}$ for some r only depending on the fixed c . Thus, $\mathfrak{p} \mapsto \chi_\mathfrak{p}(c)$ yields a character of finite order and, extending multiplicatively on the ideals coprime to some conductor $\mathfrak{f}$ indicates that $\mathfrak{p} \mapsto \chi_\mathfrak{p}(c)$ is a Hecke character. $\square$

One may then use the above definition to recover Weil's original Jacobi sum with the following observation. We set $\theta_0 = \{s/m\} + \sum_{i=1}^r \{a_i/m\}$ and note that the addition of $r + 1$ trivial terms in the sum or, more explicitly, $\theta_0 = \{s/m\} + (r + 1)\{0\} + \sum_{i=1}^r \{a_i/m\}$, contributes $r + 1$ trivial Gauss sums to the product in 3.3.5, which were demonstrated to be -1 as per 3.2.3. As such, this addition to θ_0 corresponds with multiplication of the Jacobi sum in 3.3.5 by $(-1)^{r+1}$. Thus:

$$(3.3.13) \quad J_{\theta_0}(1)(\mathfrak{p}) = J_{\underline{a}}(\mathfrak{p}).$$

3.4. Jacobi Sums are Hecke Grossencharacters

The final task before commencing the proof of Weil's is to introduce the Stickelberger element, state Stickelberger's theorem, and allude to some significant results which may play a great role within the proof of the main theorem. The former points will allow us to obtain a decomposition of $J_{\theta}(\mathfrak{p})$ and thus aid in demonstrating that J_{θ} is a Grossencharacter. For the sake of brevity, its proof will be omitted from this section. However, an explicit version of the proof of Stickelberger's theorem may be found in Washington's book *Introduction to Cyclotomic Fields* [28, p. 94]. The following definitions and statements pertaining to Stickelberger are adapted from the same text.

Definition 3.4.1. (*The Stickelberger element and ideal*) Set $G = \text{Gal}(\mathbb{Q}(\zeta_m)|\mathbb{Q})$ and denote by $\sigma_a \in G$ the automorphism which maps $\zeta_m \mapsto \zeta_m^a$ (with $(a, m) = 1$). Moreover, for $\lambda \in \mathbb{R}$ we set $\{\lambda\}$ to be the fractional part of λ . The Stickelberger element θ is defined as

$$\theta = \sum_{t \pmod{m}} \left\{ \frac{t}{m} \right\} \sigma_{-t}^{-1} \in \mathbb{Q}[G]$$

where we sum over all $t \pmod{m}$ given $(t, m) = 1$. Moreover, the Stickelberger ideal $\mathfrak{v}$ is defined to be the $\mathbb{Z}[G]$ -multiples of θ whose coefficients are integral or, more concisely, $\mathfrak{v} = \mathbb{Z}[G] \cap \theta \mathbb{Z}[G]$ [28, p. 93].

Remark 3.4.2. This definition of the Stickelberger element differs slightly from its use in Weil's paper [31, p. 489], where it is defined instead as a family of elements $\{\theta_a\}_{a \in \mathbb{Z}/m\mathbb{Z}}$ where $\theta_a = \sum_{t \pmod{m}} \left\{ \frac{ta}{m} \right\} \sigma_{-t}^{-1}$. In this section, we introduce a slight variation of Weil's convention by setting θ_{y_i} to be the Stickelberger element corresponding to the Gauss sum $g_{y_i}(\mathfrak{p})^{n_i}$, where the variable a is essentially chosen to be the numerator of $\{y_i\} \in \mathbb{Q}$.

Theorem 3.4.3. (Stickelberger) Let $\mathfrak{f}$ be a fractional ideal of $\mathbb{Q}(\zeta_m)$ and consider $\beta \in \mathbb{Z}[G]$ such that $\beta\theta \in \mathbb{Z}[G]$. Then $\mathfrak{f}^{\beta\theta}$ is principal. As such, the Stickelberger ideal annihilates the ideal class group of $\mathbb{Q}(\zeta_m)$.

Proof. See Washington [28, p. 95-98] for an explicit proof. □

We now begin the proof of the main theorem regarding these formal Jacobi sums being Hecke Grossencharacters.

Theorem 3.4.4. For each nonzero $\theta \in \mathbb{Z}[\mathbb{Q}/\mathbb{Z}]$, the function J_θ is a Hecke Grossencharacter on $I_{\mathbb{Q}(\zeta_m)}((m^2))$, where m is the lowest common multiple of the denominators of the y_i terms in θ .

Proof. By Definitions 3.2.5 and 3.3.6, given a nontrivial sum of defining data $\theta = \sum_i n_i \{y_i\} \in \mathbb{Z}[\mathbb{Q}/\mathbb{Z}]$, the quantity $J_\theta(\mathfrak{p})$ defines a well-defined multiplicative character on $I((m^2))$, obtained by first specifying its values on the primes $\mathfrak{p} \in I((m^2))$, then extending multiplicatively to the entire group.

As a prerequisite to analysing the behaviour of J_θ on principal ideals $(\alpha) \pmod{(m^2)}$, we will compute $|J_\theta(\alpha)|^2$. A well known result on the absolute value of Gauss sums (see: [13, p. 92]) states that $|g_{y_i}(\mathfrak{p})^{n_i}|^2 = \text{Nm}(\mathfrak{p})^{n_i}$. Applying this first for prime ideals $\mathfrak{p} \in I((m^2))$, and then for any $\alpha \in I((m^2))$ after extending multiplicatively, we have

$$(3.4.5) \quad |J_\theta(\alpha)|^2 = \text{Nm}(\alpha)^{\sum_i n_i}.$$

The above also holds for all conjugates of $J_\theta(\alpha)$ as for $\sigma_t \in \text{Gal}(K|\mathbb{Q})$, $J_\theta(\alpha)^{\sigma_t} = \prod_i g_{ty_i}(\mathfrak{p})^{n_i}$. Given none of the Gauss sum factors in the Jacobi sum product vanish under conjugation via σ_a , the absolute value must remain invariant under $(\cdot)^{\sigma_t}$.

We now make use of the Stickelberger relations to obtain a prime ideal decomposition of $J_\theta(\mathfrak{p})$, which is then extended to a decomposition for all $\alpha \in I((m^2))$ multiplicatively. As the key to this decomposition lies within a famous result regarding Gauss sums, and to ensure that each ideal $(g_{y_i}(\mathfrak{p})^{n_i}) \subseteq \mathbb{Z}[\zeta_m]$, rather than potentially lying in some larger ring of integers, we consider instead:

$$(J_\theta(\mathfrak{p})^m) = \prod_{i=1}^r (g_{y_i}(\mathfrak{p})^{mn_i}).$$

Then, due to a result by Stickelberger, each ideal $(g_{y_i}(\mathfrak{p})^{mn_i})$ appearing in the product $(J_\theta(\mathfrak{p})^m)$ decomposes as:

$$(3.4.6) \quad (g_{y_i}(\mathfrak{p})^{mn_i}) = \mathfrak{p}^{mn_i \theta_{y_i}}.$$

By setting $\kappa(\theta) := \sum_{i=1}^r n_i \theta_{y_i}$, 3.4.6 indicates that, for prime ideals $\mathfrak{p}$, $(J_\theta(\mathfrak{p}))$ may be decomposed as

$$(3.4.7) \quad (J_\theta(\mathfrak{p})) = \mathfrak{p}^{\kappa(\theta)}$$

where one may extend 3.4.7 multiplicatively to all $\mathfrak{a} \in I((m^2))$, thus yielding the desired result.

Now equipped with the decomposition of our Jacobi sum into prime ideals, we wish to analyse the behaviour of J_θ on principal ideals (α) such that $\alpha \equiv 1 \pmod{(m^2)}$. By first applying 3.4.7, we note that there exists a unit $u(\alpha) \in \mathcal{O}_K^\times$ such that $\alpha^{\kappa(\theta)} u(\alpha) = J_\theta((\alpha))$. Applying the fact that $\alpha \equiv 1 \pmod{(m^2)}$, we have that $J_\theta((\alpha)) \equiv u(\alpha) \pmod{(m^2)}$. One may also note that, as a consequence of 3.4.5, $|u(\alpha)| = 1$ and $|u(\alpha)^{\sigma_i}| = 1$ for all $\sigma_i \in \text{Gal}(K|\mathbb{Q})$. Then a theorem by Kronecker [16, p.173-175] allows us to conclude that $u(\alpha)$ must be a root of unity.

Indeed, $u : (1 + (m^2)\mathcal{O}_K) \rightarrow \mu_K$ may be viewed as a group homomorphism given

$$(\alpha\beta)^{\kappa(\theta)} u(\alpha\beta) = J_\theta((\alpha)(\beta)) = J_\theta((\alpha))J_\theta((\beta)) = (\alpha)^{\kappa(\theta)}(\beta)^{\kappa(\theta)} u(\alpha)u(\beta),$$

which yields $u(\alpha\beta) = u(\alpha)u(\beta)$. We claim that u is the trivial homomorphism on $(1 + \mathfrak{f}_{\text{def}}\mathcal{O}_K)$ for some defining ideal $\mathfrak{f}_{\text{def}}$, often dividing (m^2) . Proving this claim will then yield the correct infinity type for the Jacobi sum on principal ideals (α) such that $\alpha \equiv 1 \pmod{\mathfrak{f}_{\text{def}}}$. The rest of this proof will deal with demonstrating the existence of such an ideal $\mathfrak{f}_{\text{def}}$, whilst methods of computing the defining ideal can be explored in [31] and [24].

First, consider an auxiliary ideal $\mathfrak{F} \subseteq \mathcal{O}_K$. If this choice of ideal leads to $u : (1 + \mathfrak{F}\mathcal{O}_K) \rightarrow \mu_K$ becoming a trivial homomorphism, then we are done (i.e. $\mathfrak{f}_{\text{def}} = \mathfrak{F}$). As such, suppose that the homomorphism u is not trivial on $(1 + \mathfrak{F}\mathcal{O}_K)$. Given that $\ker(u) \leq (1 + \mathfrak{F}\mathcal{O}_K)$, we define $\mathfrak{f}_{\text{def}}$ to be the smallest ideal (ordered by inclusion) dividing $\mathfrak{F}$ for which $(1 + \mathfrak{f}_{\text{def}}\mathcal{O}_K) \leq \ker(u)$. Whilst not necessarily the conductor of J_θ , this tells us that there is a defining ideal $\mathfrak{f}_{\text{def}}$ for which $J_\theta((\alpha))$ decomposes just as that of 3.1.4, thus proving that the Jacobi sum J_θ is a Hecke Grossencharacter.

□

3.5. An application of Weil's theorem

We finalise this chapter by exploring one such example where this connection between Jacobi sums and Hecke Grossencharacters is indeed a useful property. For the rest of this section, we will consider the homogeneous polynomial $f(y) = y_0^3 + y_1^3 + y_2^3$ and the variety

$$X_{\mathbb{F}_p} := \{y \in \mathbb{P}^2(\mathbb{F}_p) : f(y) = 0\},$$

often simplified to just X (over $\mathbb{P}^2(\mathbb{F}_p)$). The solutions to this equation modulo $\mathfrak{p}$ will be of greater interest when $\text{Nm}(\mathfrak{p}) \equiv 1 \pmod{3}$, as the case when $\text{Nm}(\mathfrak{p}) \equiv 2 \pmod{3}$ yields a bijection between $\mathbb{F}_p^\times$ and $(\mathbb{F}_p^\times)^3$ via the cube map $\alpha \mapsto \alpha^3$. This indicates that, for $\text{Nm}(\mathfrak{p}) \equiv 2 \pmod{3}$, $f(y) = 0$ has the same number of solutions modulo $\mathfrak{p}$ as the plane $y_0 + y_1 + y_2 = 0 \pmod{\mathfrak{p}}$, which is $\text{Nm}(\mathfrak{p}) + 1$.

For the rest of this section, we fix the field $K = \mathbb{Q}(\zeta_3)$, and begin by considering finite fields of the form $\mathbb{F}_p := \mathbb{Z}[\zeta_3]/\mathfrak{p} \cong \mathbb{F}_q$ where q is some power of a prime $p \in \mathbb{Z}_{>0}$. Denoting every extension of $\mathbb{F}_p$ as $\mathbb{F}_{p^s}$, where the $s \in \mathbb{Z}_{>0}$ corresponds to the size of the extension $s = [\mathbb{F}_{p^s} : \mathbb{F}_p]$, we set $N_s := \#X_{\mathbb{F}_{p^s}}$ for each fixed prime $\mathfrak{p}$. We begin by recalling a few definitions and theorems pertaining to the study of zeta functions and L -functions of curves.

Definition 3.5.1. *Given a hypersurface defined by the homogeneous polynomial $f(x_0, \dots, x_n) = 0$ in $\mathbb{P}^n(\mathbb{F}_p)$, and the number of points N_s (where $s = [\mathbb{F}_{p^s} : \mathbb{F}_p]$) on the projective hypersurface, the zeta function defined by f is given by:*

$$(3.5.2) \quad Z_f(T) = \exp \left(\sum_{s \geq 1} \frac{N_s T^s}{s} \right).$$

Remark 3.5.3. Setting $T = p^{-s}$, we define $\zeta(X(\mathbb{F}_p), s) := Z_f(p^{-s})$ to be the local zeta function of X at $\mathfrak{p}$.

Definition 3.5.4. *Let $\chi : \mathbb{Z} \rightarrow \mathbb{C}^1$ be a Dirichlet character modulo n . Then we define the Dirichlet L -function (associated to χ) as:*

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

If $\psi : I_K(\mathfrak{f}) \rightarrow \mathbb{C}^\times$ is a Grossencharacter, then the (Hecke) L -function (associated to ψ) is similarly given by:

$$L(s, \psi) = \sum_{\gcd(\mathfrak{a}, \mathfrak{f})=1} \frac{\psi(\mathfrak{a})}{\text{Nm}(\mathfrak{a})^s}.$$

In order to work with these zeta functions, certain results explored in Ireland and Rosen's book [13, p. 102-104] will be adapted with the notation of this paper.

Theorem 3.5.5. *Consider the projective hyperplane defined by the equation $\sum_{i=0}^r b_i x_i^m = 0$, and a finite field $\mathbb{F}_p$ where $q = \text{Nm}(\mathfrak{p})$ and $q \equiv 1 \pmod{m}$. Moreover, consider the Gauss sums $g_{a_i/m}(\mathfrak{p})$ attached to the order m characters as per 3.1.7. Finally, for $G = (\mathbb{Z}/m\mathbb{Z})^\times$, we write $\theta(\underline{a}) = \sum_{i=0}^r n_i \{a_i/m\}$, where $\underline{a} = (a_0, \dots, a_r) \in G^{r+1}$. Then the number of points on this hypersurface is given by:*

$$(3.5.6) \quad q^{r-1} + \dots + q + 1 + \sum_{\substack{\underline{a} \\ \sum_i n_i a_i/m \in \mathbb{Z}}} \chi_{\mathfrak{p}} \left(\prod_{i=0}^r b_i^{a_i} \right) J_{\theta(\underline{a})}(1)(\mathfrak{p}).$$

Proof. See [13, p. 147-148] for an explicit proof. □

Let $\mathfrak{p}$ be a prime ideal of $\mathbb{Z}[\zeta_3]$. We begin by finding the number of points on $X_{\mathbb{F}_{p^s}}$ for $s = 1$, then extend to all $s \geq 0$ with a technique which will be introduced below. Applying the theorem above yields:

$$N_1 = p + 1 + \sum_{\underline{a}} J_{\theta(\underline{a})}(\mathfrak{p})(1) = q + 1 + J_{3\{1/3\}}(1)(\mathfrak{p}) + J_{3\{2/3\}}(1)(\mathfrak{p}),$$

where expressing $\psi(\mathfrak{p}) = J_{3\{1/3\}}(1)(\mathfrak{p})$ simplifies the above as

$$(3.5.7) \quad N_1 = q + 1 + \psi(\mathfrak{p}) + \overline{\psi(\mathfrak{p})}.$$

Before continuing, we introduce a way to “lift” Gauss sums from their base field to an extension of $\mathbb{F}_p$ denoted as $\mathbb{F}_{p^s}$, notationally reflecting the degree of the extension $s = [\mathbb{F}_{p^s} : \mathbb{F}_p]$. Such a technique, known as the Hasse-Davenport relation, will allow us to compute N_s , the number of points on the hyperplane over $\mathbb{P}^2(\mathbb{F}_{p^s})$ for all $s \in \mathbb{Z}_{\geq 0}$.

Theorem 3.5.8. *Let $\chi : \mathbb{F}_p \rightarrow \mathbb{C}^1$ and $\Theta_0 : \mathbb{F}_p \rightarrow \mathbb{C}^1$ be non-trivial multiplicative and additive characters respectively. Recall the extension of Θ_0 to $\mathbb{F}_{p^s}$ given by $\Theta := \Theta_0 \circ \text{Tr}_{\mathbb{F}_{p^s} | \mathbb{F}_p}$, and define*

$$\chi' := \chi \circ \text{Nm}_{\mathbb{F}_{p^s} : \mathbb{F}_p}.$$

Then using the notation as per 3.2.3, the Hasse-Davenport relation relates the Gauss sum of its base character with its lifted character as:

$$(3.5.9) \quad g(\chi') = (-1)^{s-1} g(\chi)^s$$

Proof. See [13, p. 163-166] for a complete proof of the relation. $\square$

Remark 3.5.10. Keeping with the notation of this paper, we relate the Gauss sums over $\mathbb{F}_{p^s}$ and $\mathbb{F}_p$ (denoted by $g_{a/m}(\mathfrak{p}^s)$ and $g_{a/m}(\mathfrak{p})$ respectively) as

$$g_{a/m}(\mathfrak{p}^s) = (-1)^{s-1} g_{a/m}(\mathfrak{p})^s$$

where the multiplicative character associated with $g_{a/m}(\mathfrak{p}^s)$ is $\chi_{\mathfrak{p}} \circ \text{Nm}_{\mathbb{F}_{p^s}:\mathbb{F}_p}$.

In particular, for primes $\mathfrak{P}$ above $\mathfrak{p}$ such that $[\mathbb{F}_{\mathfrak{P}} \mid \mathbb{F}_p] = s$, we observe that $\psi(\mathfrak{P}) = g_{1/3}(\mathfrak{p}^s)^3 = (-1)^{3s-3} g(\mathfrak{p})^{3s} = -(-\psi(\mathfrak{p}))^s$. Now equipped with the Hasse-Davenport relation, the point count from 3.5.7 can be extended to any extension $\mathbb{F}_{p^s}$ of $\mathbb{F}_p$ as:

$$N_s = q^s + 1 - (-\psi(\mathfrak{p}))^s - (-\overline{\psi(\mathfrak{p})})^s.$$

Substituting the above into the generating sum from 3.5.2, we obtain:

$$\sum_{s \geq 1} \left(\frac{(qT)^s}{s} + \frac{T^s}{s} - \frac{(-\psi(\mathfrak{p})T)^s}{s} - \frac{(-\overline{\psi(\mathfrak{p})}T)^s}{s} \right) = \log \left(\frac{(1 + \psi(\mathfrak{p})T)(1 + \overline{\psi(\mathfrak{p})}T)}{(1 - T)(1 - qT)} \right).$$

Exponentiating both sides then yields:

$$Z_f(T) = \frac{(1 + \psi(\mathfrak{p})T)(1 + \overline{\psi(\mathfrak{p})}T)}{(1 - T)(1 - pT)} = \frac{1 + a_p T + pT^2}{(1 - T)(1 - pT)}.$$

The term a_p is a sequence of integers depending on the primes $p \in \mathbb{Z}_{>0}$. Indeed, a_p is an integer due to the fact that it is twice the real part of $\psi(\mathfrak{p}) \in \mathbb{Z}[\zeta_3]$.

Due to 3.4.5, we also observe that $|\psi(\mathfrak{p})|^2 = p$, where the Jacobi sum $\psi(\mathfrak{p})$ also may be expressed as $x + y\zeta_3$, where $y \equiv 0 \pmod{3}$ and $x \equiv -1 \pmod{3}$ [13, p. 96]. As a consequence, for such a prime p , there exist integers $a_p = 2x - y$, $b_p = y/3$ such that:

$$4p = a_p^2 + 27b_p^2$$

where $a_p \equiv 1 \pmod{3}$. Requiring this congruence to hold then uniquely determines $2\Re(\psi(\mathfrak{p})) = a_p$, ridding any ambiguity of taking the real part of any associates of $\psi(\mathfrak{p})$. For a further explanation of these properties, one may consider [13, p.92-97].

Now, by making the substitution $T \leftrightarrow p^{-s}$, we then obtain the local zeta function of $X(\mathbb{F}_p)$:

$$(3.5.11) \quad \zeta(X(\mathbb{F}_p), s) = Z_f(p^{-s}) = \frac{1 + a_p p^{-s} + p^{1-2s}}{(1 - p^{-s})(1 - p^{1-s})}.$$

Before delving into the notion of a global zeta function, we must note that we have only defined a local zeta function at what is known as a “good” prime. This notion typically deals with primes which do not divide the discriminant of a curve, or if the Jacobian does not vanish. The “bad” primes, in this instance, are when these properties do not hold. In this case, the bad primes p are those satisfying $p \mid 3$. A quick computation in Magma tells us that the local zeta function at $p \mid 3$ is given by:

$$\zeta(X(\mathbb{F}_p), s) = \frac{1}{(1 - p^{-s})(1 - p^{1-s})}.$$

Then a global zeta function is constructed by taking the product of the local functions at all primes p :

$$(3.5.12) \quad \zeta(X, s) := \prod_p \zeta(X(\mathbb{F}_p), s).$$

Recalling that the Riemann zeta function $\zeta(s) := \sum_{n \geq 1} 1/n^s$ factorises as:

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}$$

and defining $L(X, s)$ as

$$(3.5.13) \quad L(X, s) := \prod_{p \text{ good}} \frac{1}{1 + a_p p^{-s} + p^{1-2s}} = \prod_{p \nmid f_{\text{def}}} \frac{1}{(1 + \psi(p) \text{Nm}(p)^{-s})} \frac{1}{(1 + \overline{\psi}(p) \text{Nm}(p)^{-s})},$$

where f_{def} is as per 3.4.3, we see that $\zeta(X, s)$ decomposes as $\zeta(X, s) = \zeta(s)\zeta(s-1)L(X, s)^{-1}$. Given the above proof that Jacobi sums are Hecke Grossencharacters, this then highlights the fact that our original point count can now be analysed globally by considering the Hecke L -functions $L(\psi, s)$ which arise from the point count:

$$(3.5.14) \quad L(s, \psi) = \sum_{(\mathfrak{a}, f_{\text{def}})=1} \frac{\psi(\mathfrak{a})}{\text{Nm}(\mathfrak{a})^s} = \prod_p \frac{1}{(1 + \psi(p) \text{Nm}(p)^{-s})}.$$

Here, we take f_{def} to be the conductor of the Grossencharacter, which can often be computed explicitly in Magma.

The expression in (3.5.14), often referred to as a Hecke L -function (of ψ), encapsulates the arithmetic data of the Grossencharacter (and, by extension, the variety $X_{\mathbb{F}_p}$) in an analytic form.

Hecke's work on Grossencharacters included proving that $L(s, \psi)$ admits analytic continuation and a functional equation [5, p. 101], transforming local arithmetic objects, such as point counts, into these global analytic ones.

CHAPTER 4

Descent

Given a field K , an extension L of K , and an object A over K (possibly a vector space, an algebra or a morphism), a common object of study is the behaviour of A over an extension such as $L \mid K$. We denote such an object as A_L , which is generally obtained by informally “tensoring up” A to L or, more formally, by extending its scalars to L via the identification $A_L \cong A \otimes_K L$. Such a process is considered to be, in general, a fairly manageable task. A more interesting question, and a precursor to what this chapter will focus on, relates with “undoing” this process of extension, and asking ourselves whether Grossencharacters defined over a field L came from a subfield $K \subseteq L$.

4.1. Definitions and motivations

Before defining the notion of descent for Grossencharacters, we will briefly explore an example pertaining to field extensions. Let $L = \mathbb{Q}(\zeta_5)$ and consider $\sqrt{5} = 1 + 2(\zeta_5 + \zeta_5^{-1}) \in \mathbb{Q}(\zeta_5)$. A natural question to ask is whether the Galois group can detect the smallest field $K \subseteq \mathbb{Q}(\zeta_5)$ in which $\sqrt{5}$ resides. Letting $\sigma_k \in \text{Gal}(\mathbb{Q}(\zeta_m) \mid \mathbb{Q})$ be the map satisfying $\sigma_k(\zeta_5) = \zeta_5^k$, and noting that $\text{Gal}(\mathbb{Q}(\zeta_m) \mid \mathbb{Q}) = \{\sigma_1, \sigma_2, \sigma_3, \sigma_4\}$, we observe that $\sigma_k(\sqrt{5}) = \sqrt{5}$ for $k = 1, 4$. Thus, $\sqrt{5} \in \mathbb{Q}(\zeta_5)^{\{\sigma_1, \sigma_4\}} \cong \mathbb{Q}(\sqrt{5})$. So, whilst we originally considered $\sqrt{5} \in \mathbb{Q}(\zeta_5)$, the number evidently “descends” to the quadratic subfield $\mathbb{Q}(\sqrt{5})$.

Let $I_K(\mathfrak{f})$ be the fractional ideals of K coprime to some $\mathfrak{f}$, and $L \mid K$ an extension of number fields. For a Hecke Grossencharacter $\psi : I_K(\mathfrak{f}) \rightarrow \mathbb{C}^\times$, an analogy to the extension of scalars above is to “lift” a Grossencharacter to L via composition with the ideal norm $\text{Nm}_{L \mid K}$. In other words, the character:

$$(4.1.1) \quad \Psi : I_L(\mathfrak{f}) \rightarrow \mathbb{C}^\times, \quad \Psi := \psi \circ \text{Nm}_{L \mid K},$$

becomes a Hecke character on the fractional ideals of L coprime to $\mathfrak{f}$. This leads us to define the process of Galois descent for a Grossencharacter in the following way.

Definition 4.1.2. Let $\Psi : I_L(\mathfrak{f}) \rightarrow \mathbb{C}^\times$ be a Hecke Grossencharacter on the ideals of some number field L coprime to $\mathfrak{f} \subseteq \mathcal{O}_L$. Moreover, let $H \leq \text{Aut}(L \mid \mathbb{Q})$ be the subgroup of all automorphisms $\sigma \in \text{Aut}(L \mid \mathbb{Q})$ which satisfy:

$$(4.1.3) \quad \Psi^\sigma(\mathfrak{a}) := \Psi(\sigma(\mathfrak{a})) = \Psi(\mathfrak{a}).$$

Then, we refer to the field of moduli K as the field satisfying $K = L^H$.

Remark 4.1.4. Often we will be working within a Galois extension K of $\mathbb{Q}$, and will thus write $\text{Aut}(K \mid \mathbb{Q})$ as $\text{Gal}(K \mid \mathbb{Q})$.

Remark 4.1.5. We will often refer to the field L in 4.1.2 as the field of definition. However, as per the previous discussion on lifting Grossencharacters, this field of definition need not be unique (4.1.1). In the case of Jacobi sums J_θ , the natural choice of the field of definition is $\mathbb{Q}(\zeta_m)$, with $m \in \mathbb{Z}_{>0}$ the lowest common multiple of the denominators of the defining data θ .

Definition 4.1.6. A Hecke Grossencharacter Ψ as per 4.1.2 is said to descend to its field of moduli if it can be defined as a Grossencharacter $\psi : I_K(\mathfrak{f}) \rightarrow \mathbb{C}^\times$ on the ideals of K coprime to $\mathfrak{f}$. Equivalently, Ψ descends to ψ if it factors through the norm: $\Psi = \psi \circ \text{Nm}_{L \mid K}$.

Example 4.1.7. Let ψ be a Grossencharacter and consider the map given by $\Psi = \psi \circ \text{Nm}_{L \mid K}$. Letting $H := \text{Aut}(L \mid K)$, we see that, for any $\tau \in H$ and $\mathfrak{P} \in I_L(\mathfrak{f})$:

$$\text{Nm}_{L \mid K}(\tau(\mathfrak{P})) = \prod_{\sigma \in \text{Aut}(L \mid K)} \sigma\tau(\mathfrak{P}) = \prod_{\rho \in \text{Aut}(L \mid K)} \rho(\mathfrak{P}) = \text{Nm}_{L \mid K}(\mathfrak{P}).$$

After extending multiplicatively to all $\mathfrak{a} \in I_L(\mathfrak{f})$, we have that $\Psi(\tau(\mathfrak{a})) = \Psi(\mathfrak{a})$ for all $\tau \in H$. Thus, our “lifted” Hecke character does indeed descend to the field in which it was originally defined, demonstrating that the given definition of descent above undoes this process of “lifting”.

4.2. Galois actions on J_θ

We consider the Jacobi sum as per 3.3.6 and the multiset $S = \{a_i/m\}_{1 \leq i \leq r}$, where $m \in \mathbb{Z}_{>0}$ and $a_i \in \mathbb{Z}/m\mathbb{Z}$. As before (see: Definition 3.3.6), we require that the projected sum of the elements in S , denoted by $\theta = \sum_i n_i \{a_i/m\} \in \mathbb{Q}/\mathbb{Z}$, is zero, where the n_i terms in the sum reflect the repetition of the elements in S . Then, forming a Jacobi sum requires constructing an element in $\mathbb{Q}(\zeta_m)$, the multiplicative character $\chi_p(x)$, and an additive character $\Theta(x) \in \mathbb{Q}(\zeta_p)$.

Given our assumption that $p \nmid m$, the Galois group corresponding to the field of definition of J_θ is given by $\text{Gal}(\mathbb{Q}(\zeta_m, \zeta_p) | \mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/p\mathbb{Z})^\times \cong (\mathbb{Z}/mp\mathbb{Z})^\times$. We identify elements of this Galois group with the automorphism $\sigma_{(k,\ell)} = \sigma_{(k,1)}\sigma_{(1,\ell)}$, where $\sigma_{(k,1)}$ sends m^{th} roots of unity to their k^{th} power, and $\sigma_{(1,\ell)}$ sends p^{th} roots of unity to their ℓ^{th} power. Lastly, for a general cyclotomic extension $\mathbb{Q}(\zeta_m) | \mathbb{Q}$, elements of its Galois group σ_k will be given by the restriction of $\sigma_{(k,1)}$ to $\mathbb{Q}(\zeta_m)$.

Our first objective is to show that the value of J_θ is independent of the choice of additive character Θ corresponding to the Gauss sums in its products. As such, we recall that $\Theta(x) = e^{2\pi i \text{Tr}_{\mathbb{F}_p | \mathbb{F}_p}(x)/p}$. We define a family of additive characters $\Theta^\ell(x) = e^{2\pi i \ell \text{Tr}_{\mathbb{F}_p | \mathbb{F}_p}(x)/p}$ for $\ell \in \mathbb{Z}$.

Theorem 4.2.1. *The value Jacobi sum $J_\theta(\mathfrak{p})$, with $\theta = \sum_i n_i \{a_i/m\}$ as above, is independent of the choice of additive character $\Theta^\ell(x)$.*

Proof. Let $J_\theta(\mathfrak{p})$ correspond with the product of Gauss sums whose additive character is given by $\Theta_1(x) = \Theta(x)$, and $J_{\theta,\ell}(\mathfrak{p})$ the Jacobi sum corresponding with the additive character $\Theta^\ell(x)$. Consider an arbitrary Gauss sum $g_{a_i/m,\ell}(\mathfrak{p}) = \sum_x \chi_{\mathfrak{p}}(x)^{a_i} \Theta^\ell(x)$ in the product of $J_{\theta,\ell}(\mathfrak{p})$. Then, via a substitution $\ell^{-1}z = x$, we find that

$$\begin{aligned} g_{a_i/m,\ell}(\mathfrak{p}) &= \sum_x \chi_{\mathfrak{p}}(x)^{a_i} \Theta^\ell(x) = \sum_x \chi_{\mathfrak{p}}(x)^{a_i} \Theta(\ell x) \\ &= \sum_z \chi_{\mathfrak{p}}(\ell^{-1}z)^{a_i} \Theta(z) \\ &= \chi_{\mathfrak{p}}(\ell)^{-a_i} g_{a_i/m}(\mathfrak{p}). \end{aligned}$$

Thus, the Jacobi sum $J_{\theta,\ell}(\mathfrak{p})$ is given by

$$(4.2.2) \quad J_{\theta,\ell}(\mathfrak{p}) = \chi_{\mathfrak{p}}(\ell)^{-\sum_i n_i a_i} J_\theta(\mathfrak{p}).$$

A requirement in the defining our Jacobi sum as per 3.3.6 is that $\theta \in \mathbb{Z}[\mathbb{Q}/\mathbb{Z}]$ sums to an integer. This condition will aid us in showing that the value of $J_\theta(\mathfrak{p})$ is independent of the choice of additive character in the following way. Given $\theta = \sum_i n_i \{ \frac{a_i}{m} \} \in \mathbb{Z}$, we deduce that m must divide the sum $\sum_i n_i a_i$. As such, we can express the sum in the exponent of (4.2.2) as $-\sum_i n_i a_i = mt$, for some $t \in \mathbb{Z}$ coprime to m . Then, by the construction of the character $\chi_{\mathfrak{p}}(x)$:

$$(4.2.3) \quad \chi_{\mathfrak{p}}(x)^{mt} = \left(x^{\frac{q-1}{m}} \right)^{mt} \pmod{\mathfrak{p}} \equiv 1 \pmod{\mathfrak{p}}$$

after an application of Fermat's little theorem. Hence, $J_{\theta, \ell}(\mathfrak{p}) = J_{\theta}(\mathfrak{p})$ for all ℓ coprime to p , and the value of J_{θ} is independent of the choice of additive character, thus completing the proof. $\square$

Corollary 4.2.4. *Given the Jacobi sum J_{θ} defined in $\mathbb{Q}(\zeta_m, \zeta_p)$ as per 4.2.1, the value of $J_{\theta}(\mathfrak{p})$ lies within $\mathbb{Q}(\zeta_m)$.*

Proof. Given $\sigma_{(1, \ell)} \in \text{Gal}(\mathbb{Q}(\zeta_m, \zeta_p)|\mathbb{Q})$, an automorphism only acting on primitive p^{th} -roots of unity, the previous lemma indicates that:

$$(4.2.5) \quad \sigma_{(1, \ell)}(g_{a_i/m}(\mathfrak{p})) = \sigma_{(1, \ell)}\left(\sum_x \chi_{\mathfrak{p}}(x)\Theta(x)\right) = \sum_x \chi_{\mathfrak{p}}(x)\Theta^{\ell}(x).$$

Theorem 4.2.1 then indicates that J_{θ} is invariant under conjugation of $\mathfrak{p}$ by all automorphisms of the form $\sigma_{(1, \ell)}$. As such, $\sigma_{(1, \ell)}(J_{\theta}(\mathfrak{p})) = J_{\theta}(\mathfrak{p})$, thus completing the proof. $\square$

Equipped with this independence of the choice of additive character, we set out to determine exactly how the Galois group acts on Jacobi sums, and which elements fix their Grossencharacters: $J_{\theta}(\sigma(\mathfrak{a})) = J_{\theta}(\mathfrak{a})$. Given that J_{θ} is defined on prime ideals, then extended multiplicatively to all of $I((m^2))$, we begin by first observing how $\text{Gal}(\mathbb{Q}(\zeta_m)|\mathbb{Q})$ acts on Gauss sums, then on Jacobi sums at prime ideals $\mathfrak{p} \in I((m^2))$, and lastly on all ideals after extending multiplicatively. We investigate the first part of this plan with a lemma.

Lemma 4.2.6. *Let $\sigma = \sigma_k \in \text{Gal}(\mathbb{Q}(\zeta_m, \zeta_p)|\mathbb{Q})$. Then, given the Gauss sum $g_{a_i/m}(\mathfrak{p}) = \sum_x \chi_{\mathfrak{p}}(x)^{a_i}\Theta(x)$, we have:*

$$(4.2.7) \quad g_{a_i/m}(\sigma(\mathfrak{p})) = \sigma(g_{a_i/m}(\mathfrak{p})).$$

Proof. As per the definition of $\chi_{\mathfrak{p}}(x)$ in 3.1.7, and its properties outlined in 3.1.8 and 3.1.9, this multiplicative character has the property:

$$\chi_{\sigma(\mathfrak{p})}(\sigma(x)) = \sigma(\chi_{\mathfrak{p}}(x)).$$

Consequently, taking the Gauss sum of $\sigma(\mathfrak{p})$ yields:

$$g_{a_i/m}(\sigma(\mathfrak{p})) = \sum_{x \in \mathbb{F}_{\mathfrak{p}}} \chi_{\sigma(\mathfrak{p})}(\sigma(x))^{a_i}\Theta(\sigma(x)) = \sigma\left(\sum_{x \in \mathbb{F}_{\mathfrak{p}}} \chi_{\mathfrak{p}}(x)^{a_i}\Theta(x)\right) = \sigma(g_{a_i/m}(\mathfrak{p})),$$

thus completing the proof. $\square$

A consequence of this lemma, we may identify an action on the Jacobi sum J_θ^σ for some $\sigma \in \text{Gal}(\mathbb{Q}(\zeta_m)|\mathbb{Q})$ with:

$$J_\theta(\sigma(\mathfrak{p})) = \prod_i g_{a_i/m}(\sigma(\mathfrak{p}))^{n_i} = \sigma \left(\prod_i g_{a_i/m}(\mathfrak{p})^{n_i} \right) = \sigma(J_\theta(\mathfrak{p})).$$

The theorem and corollary above indicate that we can analyse the descent of Jacobi sums J_θ merely by considering the automorphisms σ from $\text{Gal}(\mathbb{Q}(\zeta_m)|\mathbb{Q})$ which satisfy $J_\theta(\sigma(\mathfrak{a})) = J_\theta(\mathfrak{a})$. The next portion of this section will be dedicated to showing explicitly how this Galois group acts on Jacobi sums, in preparation for the main theorem of this chapter. We begin by introducing a natural “scaling” operation on the elements of $\mathbb{Z}[\mathbb{Q}/\mathbb{Z}]$ as described in [30, p. 4].

Definition 4.2.8. Let $\theta = \sum_i n_i \{y_i\} \in \mathbb{Q}/\mathbb{Z}$. We define a scaling operation $\lambda \circ \theta$, for $\lambda \in \mathbb{Z}$, given by $\lambda \circ \theta := \sum_i n_i \{\lambda y_i\}$.

Whilst we consider $\lambda \in \mathbb{Z}$ in the definition above, in our current setup we may implicitly consider λ to be an element of $\mathbb{Z}/m\mathbb{Z}$. Moreover, when it is clear, we will often drop the symbol and just write $\lambda\theta = \lambda \circ \theta$.

Lemma 4.2.9. Given a Jacobi sum $J_\theta(\mathfrak{p}) = \prod_i g_{a_i/m}(\mathfrak{p})^{n_i}$ and $\sigma_k \in \text{Gal}(\mathbb{Q}(\zeta_m)|\mathbb{Q})$, we obtain the following identity:

$$J_\theta(\sigma_k(\mathfrak{p})) = J_{k \circ \theta}(\mathfrak{p}).$$

Proof. We first consider an arbitrary Gauss sum in the Jacobi sum product $g_{a_i/m}(\mathfrak{p}) = \sum_x \chi_{\mathfrak{p}}(x)^{a_i} \Theta(x)$. Given that $p \nmid m$, we see that

$$(4.2.10) \quad \sigma_k(g_{a_i/m}(\mathfrak{p})) = \sigma_k \left(\sum_x \chi_{\mathfrak{p}}(x)^{a_i} \Theta(x) \right) = \sum_x \chi_{\mathfrak{p}}(x)^{ka_i} \Theta(x) = g_{k(a_i/m)}(\mathfrak{p}).$$

Then, by Definition 3.3.6, we have

$$(4.2.11) \quad \sigma_k(J_\theta(\mathfrak{p})) = \sigma_k \left(\prod_i g_{a_i/m}(\mathfrak{p})^{n_i} \right) = \prod_i g_{k(a_i/m)}(\mathfrak{p})^{n_i} = J_{k \circ \theta}(\mathfrak{p}),$$

thus completing the proof. $\square$

Lemma 4.2.9 allows us to re-frame the Galois descent of Jacobi sums in the following manner. Given the canonical isomorphism $\varphi : \text{Gal}(\mathbb{Q}(\zeta_m)|\mathbb{Q}) \rightarrow (\mathbb{Z}/m\mathbb{Z})^\times$, we may then deduce that the subgroup $\varphi(H) \leq (\mathbb{Z}/m\mathbb{Z})^\times$ is precisely the set of integers $k \in (\mathbb{Z}/m\mathbb{Z})^\times$ which leave the set

S fixed after scaling (i.e. $kS = S$). Thus, unless otherwise specified, we identify the set H of the automorphisms which leave J_θ fixed as precisely the set of these integers which leave our set S defining the Jacobi sum fixed. A brief computation in Magma gives us the following examples demonstrating this phenomenon.

Example 4.2.12. Consider the Jacobi sum $J_{\{2/5\}+\{3/5\}}(\mathfrak{p})$. Computing in Magma tells us that the field of definition of $J_{\{2/5\}+\{3/5\}}(\mathfrak{p})$ is $\mathbb{Q}(\sqrt{5})$. Then, H is obtained by noting that, for $-1 \in \mathbb{Z}/5\mathbb{Z}$:

$$(-1)S = (-1) \left\{ \frac{2}{5}, \frac{3}{5} \right\} = \left\{ \frac{-2}{5}, \frac{-3}{5} \right\} \equiv \left\{ \frac{3}{5}, \frac{2}{5} \right\} = S,$$

and observing that no other element apart from 1 permutes the elements in S . Thus, $H = \{1, 4\}$ and $K = \mathbb{Q}(\zeta_5)^H = \mathbb{Q}(\sqrt{5})$, the same result as the computation in Magma.

Example 4.2.13. Our next example comes from [30, p. 5] and considers the Grossencharacter $J_{\{5/8\}+\{7/8\}-\{4/8\}}(\mathfrak{p})$. It is indicated by Magma that this Grossencharacter is defined over the field $\mathbb{Q}/(x^2 + 2) \cong \mathbb{Q}(\sqrt{-2})$. We can quickly verify as well that $H = \{1, 3\}$, meaning that

$$K = \mathbb{Q}(\zeta_8)^H = \mathbb{Q}(\sqrt{-2}).$$

4.3. Descent of Jacobi sums and Hecke Grossencharacters

The theorem which this section concerns itself with intends to relate the Jacobi Grossencharacters initially defined over $\mathbb{Q}(\zeta_m)$ with another simplified Grossencharacter defined over its field of moduli. After computing the field of moduli with the previously discussed method, we aim to prove that our new Grossencharacter is well-defined on the set of fractional ideals of K coprime to (m^2) . Then, we will finalise this chapter with an exploration of a far simpler method of proving this theorem by considering idelic Hecke Grossencharacters.

Let $L = \mathbb{Q}(\zeta_m)$. Given a Jacobi sum $J_\theta(\mathfrak{p})$ initially constructed over the group of ideals $I_L((m^2))$ with data $S = \{a_i/m\}_{1 \leq i \leq r}$, we wish to show that this Hecke character descends to a new simplified function on $I_K((m^2))$, with $K = L^H$. In order to distinguish these functions, we denote a “descended” Jacobi sum as $J_{0,\theta} : I_K((m^2)) \rightarrow \mathbb{C}^\times$. A Jacobi sum “pre-descent” will continue to be written as J_θ .

We propose that this new function $J_{0,\theta} : I_K((m^2)) \rightarrow \mathbb{C}^\times$ can be constructed multiplicatively as such: for any prime ideal $\mathfrak{p} \in \mathcal{O}_K$, we set:

$$(4.3.1) \quad J_{0,\theta}(\mathfrak{p}) := J_\theta(\mathfrak{P}\mathcal{O}_L),$$

where $\mathfrak{P} \cap \mathcal{O}_K = \mathfrak{p}$. Indeed, this restriction is independent of the choice of $\mathfrak{P}$ above $\mathfrak{p}$ given that elements of the Galois group act transitively on the primes in the factorisation $\mathfrak{p}\mathcal{O}_K = \prod_{i=1}^g \mathfrak{P}_i^{e_i}$. More explicitly, given the choice of another prime $\mathfrak{P}'$ where $J_{0,\theta}(\mathfrak{p}) = J_\theta(\mathfrak{P}'\mathcal{O}_L)$, there is a $\sigma \in \text{Gal}(\mathbb{Q}(\zeta_m)|\mathbb{Q})$ such that $\sigma(\mathfrak{P}') = \mathfrak{P}$, giving us:

$$J_\theta(\mathfrak{P}'\mathcal{O}_L) = J_{0,\theta}(\mathfrak{p}) = J_{0,\theta}(\sigma(\mathfrak{p})) = J_\theta(\sigma(\mathfrak{P}')\mathcal{O}_L) = J_\theta(\mathfrak{P}\mathcal{O}_L).$$

Then, given the way in which we extend Jacobi sums multiplicatively on ideals $\mathfrak{a}, \mathfrak{b} \in I_K((m^2))$, $J_{0,\theta}$ becomes a multiplicative function on the fractional ideals of K coprime to (m) . The last condition left to check is that of the behaviour of the absolute value of $J_\theta(\mathfrak{p})$ before and after descent. Recall that, as per 3.4.5, $J_\theta(\mathfrak{p})$ satisfies

$$|J_\theta(\mathfrak{p})|^2 = |\text{Nm}_{L|\mathbb{Q}}(\mathfrak{p})|^{\sum_i n_i}$$

given the finite sum $\theta = \sum_i n_i \{a_i/m\} \equiv 0 \pmod{\mathbb{Z}}$. This leads us to the following definition.

Definition 4.3.2. Given $J_\theta(\mathfrak{p})$ with $\theta = \sum_i n_i \{a_i/m\}$, we define the weight of the Jacobi sum $w(\theta)$ to be $w(\theta) := \sum_i n_i$.

Remark 4.3.3. More often than not, we need not distinguish between the weight of two different Jacobi sums. In such a case, we merely write $w = w(\theta)$.

A potential problem which may arise when considering the weight w of a Jacobi sum is that of the addition of trivial characters, such as in equation 3.3.13. In order to remedy this issue, we only allow for non-trivial contributions to the weight of a Jacobi sum. Then, our prototype descended Jacobi sum $J_{0,\theta}(\mathfrak{p})$ satisfies the norm condition above:

$$|J_{0,\theta}(\mathfrak{p})|^2 = |\text{Nm}_{K|\mathbb{Q}}(\mathfrak{p})|^w.$$

To compute $J_{0,\theta}$ on $I_L((m^2))$, we merely have to lift the character back to L via its composition with the norm:

$$|J_\theta(\mathfrak{P})|^2 = |J_{0,\theta}(\text{Nm}_{L|K}(\mathfrak{P}))|^2 = |\text{Nm}_{K|\mathbb{Q}}(\text{Nm}_{L|K}(\mathfrak{P}))|^w = |\text{Nm}_{L|\mathbb{Q}}(\mathfrak{P})|^w.$$

The purpose of specifying whether we are working in the field of moduli or the base field is to avoid certain anomalies when considering the same ideal in two different extensions. We illuminate this with an example, which will be further examined following the proof of the theorem below.

Example 4.3.4. Assuming 4.3.5 for now, consider the Grossencharacter $\psi(\mathfrak{p}) := J_{0,\theta}(\mathfrak{p})$ where $\theta = \{1/3\} + \{2/3\}$ and $w = 2$. Whilst being constructed in $L = \mathbb{Q}(\zeta_3)$, it descends to $K = \mathbb{Q}$ with $H = \{1, -1\}$. Then, computing the value of $J_{0,\theta}((5)\mathbb{Z})$ in Magma, we observe that

$$|J_{0,\theta}((5)\mathbb{Z})|^2 = 5^2 = |\mathrm{Nm}_{\mathbb{Q}|\mathbb{Q}}((5))|^2,$$

and computing the value with the ideals of $\mathbb{Z}[\zeta_3]$ correspondingly yields:

$$|J_{0,\theta}((5)\mathbb{Z}[\zeta_3])|^2 = |J_{0,\theta}((5)^{[\mathbb{Q}(\zeta_3)|\mathbb{Q}]}\mathbb{Z})|^2 = 5^4 = |\mathrm{Nm}_{\mathbb{Q}(\zeta_3)|\mathbb{Q}}((5))|^2.$$

We will now give a proof of the theorem that $J_{0,\theta}$ continues to be a Grossencharacter on its restriction post-descent.

Theorem 4.3.5. *With $H := \{k \in (\mathbb{Z}/m\mathbb{Z})^\times : kS = S\}$ given as before, the Jacobi Grossencharacter J_θ descends to the function $J_{0,\theta} : I_K((m^2)) \rightarrow \mathbb{C}^\times$, where $K = \mathbb{Q}(\zeta_m)^H$. Moreover, the restriction of J_θ to $J_{0,\theta}$ after descent preserves the Grossencharacter property of the Jacobi sum.*

Proof. The function $J_{0,\theta}$ above, by construction, is a well-defined multiplicative character on the group I_K , which means that it suffices to merely verify that its infinity type is in accordance with 3.1.4.

We claim that, given $\mathfrak{f}_{\mathrm{def}}$ as per 3.4.4, $\mathfrak{f}_0 := \mathfrak{f}_{\mathrm{def}} \cap \mathcal{O}_K$ is a defining ideal of $J_{0,\theta}$. Given that $\mathfrak{f}_0 \subseteq \mathfrak{f}_{\mathrm{def}}$, each $\alpha \in (1 + \mathfrak{f}_0\mathcal{O}_K)$ is also in $(1 + \mathfrak{f}_{\mathrm{def}}\mathcal{O}_L)$. Hence, we consider the ideal $(\alpha) \subseteq \mathcal{O}_K$ and observe that:

$$J_{0,\theta}((\alpha)) = J_\theta((\alpha)\mathcal{O}_L) \equiv u(\alpha)\alpha^{\kappa(\theta)}$$

where $u : (1 + \mathfrak{f}_0\mathcal{O}_K)^\times \rightarrow \mu_K$ and $\kappa(\theta)$ are borrowed from 3.4.4. However, given that u was the trivial homomorphism on $(1 + \mathfrak{f}_{\mathrm{def}}\mathcal{O}_L)^\times$ by construction of the ideal $\mathfrak{f}_{\mathrm{def}}$, the same must hold on the subgroup $(1 + \mathfrak{f}_0\mathcal{O}_K)^\times$. Thus:

$$J_{0,\theta}((\alpha)) = \alpha^{\kappa(\theta)},$$

meaning the infinity type of J_θ is preserved under descent to $J_{0,\theta}$. As such, $J_{0,\theta}$ obeys the multiplicativity and infinity type of a Grossencharacter, thus demonstrating that Jacobi sums descend to Hecke characters defined on their field of moduli. $\square$

Certainly, one of the greatest consequences of the above theorem lays within the computation of Jacobi sums. Whilst the previous chapter studied Jacobi sums as Grossencharacters on $I_{\mathbb{Q}(\zeta_m)}(\mathfrak{f})$ for some conductor $\mathfrak{f}$, this notion was ultimately a formal one, as computational algebra systems such as Magma are typically working with these descended Hecke characters. If one desired to input ideals from some extension of K , then example 4.3.4 provides the necessary strategy to do so. For the rest of this paper, we will identify J_θ with the descended Jacobi sum, as this will be the most important Hecke character for counting points on varieties over finite fields. If we are required to work with the Jacobi sum pre-descent, we will denote this Grossencharacter by $\tilde{J}_\theta$.

Another importance of this theorem on descended Jacobi sums may be viewed within two examples of point counts over $\mathbb{P}^1(\mathbb{F}_p)$: $X_{3,\mathbb{F}_p} = \{[x : y] \in \mathbb{P}^1(\mathbb{F}_p) : x^3 + y^3 = 0\}$ and $X_{5,\mathbb{F}_p} = \{[x : y] \in \mathbb{P}^1(\mathbb{F}_p) : x^5 + y^5 = 0\}$, where $\text{Nm}(\mathfrak{p}) \equiv 1 \pmod{m}$. The former will give rise to an associated Jacobi sum with “full descent”, meaning that it descends to a field of moduli of $\mathbb{Q}$, whilst the latter variety’s Grossencharacter has a “partial descent” to $\mathbb{Q}(\sqrt{5})$.

Example 4.3.6. Consider the variety $X_{3,\mathbb{F}_p} = \{[x : y] \in \mathbb{P}^1(\mathbb{F}_p) : x^3 + y^3 = 0\}$. By theorem 3.5.5, we have that

$$N_1 = 1 + J_{\theta(1,2)}(1)(\mathfrak{p}) + J_{\theta(2,1)}(1)(\mathfrak{p}) = 1 + 2J_{\theta(1,2)}(1)(\mathfrak{p}).$$

The descent of $J_{\theta(1,2)}(1)(\mathfrak{p})$ to a character over $\mathbb{Q}$ becomes evident within the very nature of the point-count. In particular, J_θ descends to what is known as the minimal field of definition, which is a term to describe any further descent of a Grossencharacter which were not picked up by the actions from the Galois group.

The variety $X_{3,\mathbb{F}_p}$ can be viewed as the solutions to $z^3 + 1 \equiv 0 \pmod{\mathfrak{p}}$ after setting $z = x/y$ (also following the verification that there are no solutions when x or y are zero). Then, $X_{3,\mathbb{F}_p}$ factors as $(z + 1)(z^2 - z + 1) \equiv 0 \pmod{\mathfrak{p}}$, meaning that $z = -1$ is always a solution, and we merely need to consider $z^2 - z + 1 \equiv 0 \pmod{\mathfrak{p}}$ or, in other terms, if there exists a cube root of unity in the finite field $\mathbb{F}_p$. This is encapsulated by the Dirichlet character $\chi_{-3}(p) = \left(\frac{-3}{p}\right)$, where:

$$\chi_{-3}(p) = \begin{cases} 1 & p \equiv 1 \pmod{3}; \\ -1 & p \equiv 2 \pmod{3}; \\ 0 & p \equiv 0 \pmod{3}. \end{cases}$$

We note as well that, for any degree s field extension $\mathbb{F}_{p^s} : \mathbb{F}_p$, we have that $N_s := \#X_{3,\mathbb{F}_{p^s}} = 1 + 2\chi_{-3}(p)^s$ for $p^s \equiv 1 \pmod{3}$ and $N_s = 1$ otherwise. Given that the term 1 in N_s merely contributes a factor of $1/(1-T)$ to the final zeta function, we will simplify the process by substituting $N_s \leftrightarrow (N_s - 1)$. As a result, for $p \equiv 1 \pmod{3}$, $Z_{X_3,\mathbb{F}_p}$ becomes:

$$Z_{X_3,\mathbb{F}_p}(T) = \exp\left(2 \sum_s \frac{T^s}{s}\right) = \frac{1}{(1-T)^2}$$

and for $p \equiv 2 \pmod{3}$:

$$Z_{X_3,\mathbb{F}_p}(T) = \exp\left(2 \sum_{s \text{ even}} \frac{T^s}{s}\right) = \exp(-\log(1-T^2)) = \frac{1}{1-T^2}$$

where the penultimate equality is achieved by summing

$$-\log(1-T) - \log(1+T) = \sum_s (1 + (-1)^s) \frac{T^s}{s} = 2 \sum_{s \text{ even}} \frac{T^s}{s}.$$

We combine both of these results by observing that, if we let $Z_{X_3,\mathbb{F}_p}(T) = 1/L_p(T)$, where $L_p(T)$ is a polynomial in T , then:

$$L_p(T) = \begin{cases} (1-T)(1-\chi_{-3}(p)T) & p \equiv 1, 2 \pmod{3}; \\ (1-T) & p = 3, \end{cases}$$

where the final result comes from the fact that the Jacobian of $X_{3,\mathbb{F}_p}$ vanishes when $p = 3$, rendering it a bad prime. Then:

$$\zeta(X_3, s) = \prod_p \frac{1}{L_p(p^s)} = \prod_p \frac{1}{(1-p^s)(1-\chi_{-3}(p)p^{-s})} = \zeta(s)L(s, \chi_{-3}),$$

where $L(s, \chi_{-3}) = \sum_{n \geq 1} \chi_{-3}(n)/n^s$ is the L -function associated with the Kronecker symbol $\chi_{-3}(p)$.

Example 4.3.7. We now consider the variety $X_{5,\mathbb{F}_p}$, where applying theorem 3.5.5 immediately tells us that, for $s = 1$:

$$N_1 = 1 + J_{\theta(1,4)}(1)(\mathfrak{p}) + J_{\theta(4,1)}(1)(\mathfrak{p}) + J_{\theta(2,3)}(1)(\mathfrak{p}) + J_{\theta(3,2)}(1)(\mathfrak{p})$$

which simplifies to $N_1 = 1 + 2(J_{\theta(1,4)}(1)(\mathfrak{p}) + J_{\theta(2,3)}(1)(\mathfrak{p}))$. Theorem 4.3.5 tells us that both Jacobi sums descend to $\mathbb{Q}(\sqrt{5})$ as their defining sets $S = \{1/5, 4/5\}$ and $S' = \{2/5, 3/5\}$ are both fixed via multiplication by 4. One may also note that $S' = 2S$.

We claim that these Jacobi sums are equal. If the prime $\mathfrak{p}$ of $\mathbb{Q}(\sqrt{5})$ is inert in $\mathbb{Q}(\zeta_5)$, then $\sigma\mathfrak{p} = \mathfrak{p}$ for all $\sigma \in \text{Gal}(\mathbb{Q}(\zeta_5)|\mathbb{Q}(\sqrt{5}))$, which gives us the desired result. Given any prime $\mathfrak{p}$ of $\mathbb{Q}(\sqrt{5})$ which splits or ramifies in $\mathbb{Q}(\zeta_5)$, then $\mathfrak{p}\mathcal{O}_{\mathbb{Q}(\sqrt{5})} = \mathfrak{P}\sigma_2(\mathfrak{P})$ as $\text{Gal}(\mathbb{Q}(\zeta_5)|\mathbb{Q}(\sqrt{5})) = \{1, \sigma_2\}$. By the construction of our descended Jacobi sum:

$$\begin{aligned} J_{\theta(1,4)}(1)(\mathfrak{p}\mathcal{O}_{\mathbb{Q}(\sqrt{5})}) &= \tilde{J}_{\theta(1,4)}(1)(\mathfrak{P}\mathcal{O}_{\mathbb{Q}(\zeta_5)}) = \tilde{J}_{\theta(1,4)}(1)(\sigma_2(\mathfrak{P})\mathcal{O}_{\mathbb{Q}(\zeta_5)}) \\ &= \sigma_2 \left(\tilde{J}_{\theta(1,4)}(1)(\mathfrak{P}\mathcal{O}_{\mathbb{Q}(\zeta_5)}) \right) \\ &= J_{\theta(2,3)}(1)(\mathfrak{p}), \end{aligned}$$

thus proving the claim. As a consequence, this allows us to set $\psi(\mathfrak{p}) := J_{\theta(1,4)}(1)(\mathfrak{p}) = J_{\theta(2,3)}(1)(\mathfrak{p})$. A brief computation in Magma indicates that ψ is the Hecke character of trivial infinity type given by:

$$\psi(\mathfrak{p}) = \begin{cases} 1 & \text{Nm}(\mathfrak{p}) \equiv 1 \pmod{5}; \\ -1 & \text{Nm}(\mathfrak{p}) \equiv -1 \pmod{5}. \end{cases}$$

A Grossencharacter is said to descend to its minimal field of descent whenever there is further descent which was not detected by the process outlined in 4.1.2. We claim that ψ descends further to a Grossencharacter $\chi_5(2, (p))$ on its minimal field of definition $\mathbb{Q}$. We have that $p \in \mathbb{Z}$ splits in $\mathbb{Q}(\sqrt{5})$ if and only if $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right) = 1$, meaning that $p \equiv \pm 1 \pmod{5}$. In this case, we have that $\psi(p) := \Psi(\mathfrak{p})$ where $\mathfrak{p} \mid p$. The prime p is inert in $\mathbb{Q}(\sqrt{5})$ if and only if $p \equiv \pm 2 \pmod{5}$. Given that 2 is a primitive root modulo 5, we set $\psi(p) = i$ for $p \equiv 2 \pmod{5}$, and get that

$$\chi_5(2, (p)) = \begin{cases} 1 & p \equiv 1 \pmod{5}; \\ i & p \equiv 2 \pmod{5}; \\ -i & p \equiv 3 \pmod{5}; \\ -1 & p \equiv 4 \pmod{5}. \end{cases}$$

which may also be identified as a Dirichlet character $\chi_5(2, p)$ over $\mathbb{Z}$. Now, a brief computation in Magma indicates that the zeta function $Z_{X_{5, \mathbb{F}_p}}(T)$ attached to $X_{5, \mathbb{F}_p}$ is given by:

$$Z_{X_{5, \mathbb{F}_p}}(T) = \frac{1}{(1 + \chi_5(2, p)T)(1 + \overline{\chi_5(2, p)T})},$$

leading to the global zeta function

$$\zeta(X_5, s) = \prod_p \frac{1}{(1 + \chi_5(2, p)p^{-s})(1 + \overline{\chi_5(2, p)}p^{-s})} = L(s, \chi_5(2, \cdot))L(s, \overline{\chi_5(2, \cdot)})$$

after computing $Z_{X_5, \mathbb{F}_p} = 1$ at the bad prime $p = 5$.

4.4. A view of the adelic framework

We conclude this chapter by giving a brief view of Hecke characters through the adelic framework, and the behaviour of descent after considering them in this way. The discussion of the adèle ring $\underline{K}$ of a number field K will mainly summarise the content of [27, p. 445-456], whilst the discussion of the idelic Hecke characters will follow [22] and [29].

Let K be a global field, $\nu \in \text{Pl}(K)$, and K_ν its completion at ν with the corresponding valuation ring $R_\nu := \{x \in K_\nu : |x|_\nu \leq 1\}$.

Definition 4.4.1. *The adèle ring $\underline{K}$ of K is given by the restricted direct product of K_ν (with respect to R_ν)*

$$\underline{K} := \prod'_\nu K_\nu = \left\{ (x_\nu)_\nu \in \prod_\nu K_\nu : x_\nu \in R_\nu \text{ for all but finitely many } \nu \in \text{Pl}(K) \right\}.$$

Along with the adèle ring, we have the idele group $\underline{K}^\times$, given similarly by the restricted direct product $\underline{K}^\times := \prod'_\nu K_\nu^\times := \{(x_\nu)_\nu \in \prod_\nu K_\nu^\times : |x_\nu|_\nu = 1 \text{ for all but finitely many } \nu \in \text{Pl}(K)\}$. There is an injective diagonal homomorphism $\iota : K^\times \rightarrow \underline{K}^\times$ given by $a \mapsto (a, \dots, a)$. The image $\iota(K^\times) \subseteq \underline{K}^\times$ is discrete, meaning that for every $a \in K^\times$, there is an open neighbourhood $U_a \subseteq \underline{K}^\times$ of a such that $U_a \cap \underline{K}^\times = \{\iota(a)\}$ (see [18, p. 170-171] for a proof of this fact).

Definition 4.4.2. *Defining the idele class group as the quotient $C_K := \underline{K}^\times / K^\times$, we have that an (idelic) Hecke Grossencharacter is a continuous group homomorphism $\psi : C_K \rightarrow \mathbb{C}^\times$.*

Remark 4.4.3. From any idelic Grossencharacter $\psi = \bigotimes_\nu \psi_\nu$, we can construct a classical one (on $I_K(\mathfrak{f})$) by first constructing a map $\psi_{\text{classic}} : I_K(\mathfrak{f}) \rightarrow \mathbb{C}^\times$ which agrees with the former map on primes $\mathfrak{p}_\nu$:

$$\psi_{\text{classic}}(\mathfrak{p}_\nu) := \psi_\nu(\mathfrak{m}_\nu R_\nu^\times)$$

for a uniformiser $\mathfrak{m}_\nu$ of K_ν . Extending multiplicatively should then yield a classical Hecke character with the desired infinity type.

We now state one of the big theorems of class field theory. We denote by F^{sep} the separable closure of F , and recall that an extension $F | K$ is abelian if $\text{Gal}(F|K)$ is abelian. Moreover, an open subgroup $H \leq C_F$ refers to H being open as a subgroup of the topological group C_F . Whilst vital to the study of the adelic framework, we omit certain topological details in this exploration.

Theorem 4.4.4. *There is a bijection between the sets:*

$$\{F \subseteq K \subseteq F^{\text{sep}} : K \text{ finite, abelian extension}\} \leftrightarrow \{H \leq C_F : H \text{ open and of finite index}\}$$

given by $K \mapsto F^\times \text{Nm}_{K|F}(C_K)$. Together with this bijection, we obtain the Artin isomorphism $\varphi_{K|F} : C_F / (F^\times \text{Nm}_{K|F}(C_K)) \rightarrow \text{Gal}(K|F)$ of the extension $K | F$.

Colloquially, this theorem tells us that the idele class group C_F encodes much of the information of its finite abelian extensions K with $F^{\text{sep}} | K | F$. These “norm subgroups” $F^\times \text{Nm}_{K|F}(C_K)$ are precisely these building blocks, and allow us to now view descent through the adelic framework.

Let $\Psi : C_L \rightarrow \mathbb{C}^\times$ be a Grossencharacter. Suppose that we have an extension $L | K$ such that, for all $a \in C_L$ and $\sigma \in \text{Gal}(L|K)$, $\Psi^\sigma(a) = \Psi(\sigma(a)) = \Psi(a)$. Then, Ψ is constant on $\ker(N_{L|K})$ and thus factors through $C_L / \ker(N_{L|K}) \cong N_{L|K}(C_L) \leq C_K$, defining another character $\psi : C_K \rightarrow \mathbb{C}^\times$ satisfying $\Psi = \psi \circ \text{Nm}_{L|K}$. This demonstrates that, with some more complicated machinery, the proof of theorem 4.3.5 can be further refined when considering Hecke characters through the adelic framework.

Finite Field Hypergeometric Functions

This chapter will intend to explore the complex interconnections between Jacobi sums and finite field hypergeometric functions (often abbreviated as HGMs). In the first section, we will recall the definition of the classical hypergeometric function and a bridge to its finite field analogue, with the backdrop of its interpretation from Galois representations. Then, we will outline a method to extend HGMs to a less restrictive domain by expressing them in terms of (descended) Jacobi sums.

5.1. Bridging the classical and finite field hypergeometric functions

We begin this section with a brief overview of the classic hypergeometric series coupled with an exploration of where their finite field analogue comes from. The former exploration will be a summary of various well regarded papers and expository articles on HGMs, such as [7], [8], and [9], and the latter part will recall a setup similar to [14] and connect these HGMs to representations of the absolute Galois group. We start off with recalling a few definitions and identities.

Definition 5.1.1. *The (rising) Pochhammer symbol $(x)_n$ of x is given by:*

$$(x)_n = \begin{cases} 1 & n = 1 \\ x(x+1) \dots (x+n-1) & n > 1. \end{cases}$$

Recalling the common extension of the factorial to the complex numbers $\Gamma(z)$ given by the integral $\Gamma(z) = \int_0^\infty t^z e^{-t} \frac{dt}{t}$, we may also express $(x)_n$ as

$$(x)_n = \frac{\Gamma(x+n)}{\Gamma(x)}.$$

Definition 5.1.2. *For $\Re(x) > 0$ and $\Re(y) > 0$, we define the beta function $B(x, y)$ as:*

$$B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt.$$

Remark 5.1.3. It can be shown, by considering the integral of $\Gamma(x)\Gamma(y)$ and a change of variables, that the beta function above can be expressed in terms of the gamma function:

$$(5.1.4) \quad B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

Theorem 5.1.5. (*Euler reflection formula*) For $\alpha \in \mathbb{C}$, the following identity holds

$$(5.1.6) \quad \Gamma(\alpha)\Gamma(1-\alpha) = \frac{\pi}{\sin(\pi\alpha)}.$$

Proof. After a change of variables, one may note that

$$\Gamma(\alpha)\Gamma(1-\alpha) = \int_0^\infty \int_0^\infty \frac{\xi^{\alpha-1}}{1+\xi} e^{-\nu} d\nu d\xi = \int_0^\infty \frac{\xi^{\alpha-1}}{1+\xi} d\xi = \frac{\pi}{\sin(\pi\alpha)}$$

where the final equality is computed via typical means of contour integration. □

We now introduce the notion of a classical hypergeometric function.

Definition 5.1.7. (*Classical hypergeometric function*) Let $\alpha = \{a_1, \dots, a_{n+1}\}$ and $\beta = \{b_1, \dots, b_n\}$ be sets of complex parameters. Then, the classical hypergeometric function ${}_{n+1}F_n[\alpha; \beta; z]$ is given by:

$${}_{n+1}F_n \left[\begin{matrix} a_1 & \dots & a_{n+1} \\ b_1 & \dots & b_n \end{matrix}; z \right] = \sum_{\eta=0}^{\infty} \frac{(a_1)_\eta \dots (a_{n+1})_\eta}{(b_1)_\eta \dots (b_n)_\eta} \frac{z^\eta}{\eta!}.$$

Remark 5.1.8. Certain evaluation formulas link the classical hypergeometric sum with the gamma and beta function. Examples of these formulas can be found in [9, p. 26-27], such as

$${}_2F_1 \left[\begin{matrix} a & b \\ c \end{matrix}; 1 \right] = \frac{B(b, c-a-b)}{B(b, c-b)} = \frac{\Gamma(b)\Gamma(c-a-b)/\Gamma(c-a)}{\Gamma(b)\Gamma(c-b)/\Gamma(c)} = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-b)\Gamma(c-a)}.$$

These identities will provide further analogies between the classical and finite field hypergeometric sums.

Deviating slightly from the case of the classic hypergeometric function, we introduce its finite field analogue and a few notions from the representation of Galois groups in order to exhibit where it arises from before providing a “Rosetta stone” in order to emphasise the similarities between these concepts. We recall the definition from [14, p. 14].

Definition 5.1.9. For $\mathfrak{p} \in \text{Spec}(\mathbb{Z}[\zeta_m])$ with $\text{Nm}(\mathfrak{p}) = q$, we let $\alpha, \beta \subset (1/(q-1))\mathbb{Z}$ be the multisets of hypergeometric parameters with $\#\alpha = \#\beta = n$. Then, for $t \in \mathbb{F}_{\mathfrak{p}}$, we define the (finite field period-normalised) hypergeometric sum as:

$$(5.1.10) \quad H_{\mathfrak{p}}(\alpha, \beta, t) = \frac{1}{q-1} \sum_{\mu=0}^{q-2} \chi_{\mathfrak{p}}((-1)^n t)^{\mu/(q-1)} \prod_{i=1}^n \frac{g_{\alpha_i + \mu/(q-1)}(\mathfrak{p}) g_{\beta_i + \mu/(q-1)}(\mathfrak{p})}{g_{\alpha_i - \beta_i}(\mathfrak{p})}.$$

We recall a few definitions and results from the representation of Galois groups in order to provide further context to where these HGMs arise from. For a number field K and its algebraic closure $\overline{K}$, we recall the the group of automorphisms which fix the elements of K : $\text{Aut}(\overline{K} | K) = \{\sigma \in \text{Aut}(K) \mid \sigma(\alpha) = \alpha \text{ for all } \alpha \in K\}$. Endowing $\text{Aut}(\overline{K} | K)$ with the Krull topology, where the neighbourhood base of the identity $\text{id} \in \text{Aut}(\overline{K} | K)$ is given by the sets $G(S) := \{\sigma \in \text{Aut}(\overline{K} | K) \mid \sigma(s) = s \text{ for } s \in S\}$ where $S \subseteq \overline{K}$ is a finite subset, we arrive at the absolute Galois group $G_K := \text{Aut}(\overline{K} | K)$, often writing $\text{Gal}(\overline{K}|K)$ in place of $\text{Aut}(\overline{K} | K)$. Whilst the degree of this extension will naturally be infinite, G_K does have the advantage of encoding all of the finite Galois extensions of K within it, hence it playing a great role in the study of Galois theory. For a more profound exploration of this topic, one may consider [17, p. 161-180] and [23, p. 261-265].

Definition 5.1.11. A representation ρ of $G_K = \text{Gal}(\overline{K}|K)$ is a continuous homomorphism $\rho : G_K \rightarrow \text{GL}_n(V)$, where V is a finite dimensional vector space. We say that ρ is an ℓ -adic representation when V is a vector space over $\mathbb{Q}_{\ell}$.

Example 5.1.12. Consider the algebraic closure of $\overline{\mathbb{Q}}$ and the set $\mu_n = \{\zeta \in \overline{\mathbb{Q}}^{\times} : \zeta^n = 1\}$ of n -th roots of unity. Given a prime number ℓ , $G_{\mathbb{Q}}$ acts on μ_{ℓ^n} and hence also on $T_{\ell}(\mu) := \varprojlim \mu_{\ell^n}$. Extending scalars turns $T_{\ell}(\mu)$ into a vector space $V_{\ell}(\mu) = T_{\ell}(\mu) \otimes_{\mathbb{Z}_{\ell}} \mathbb{Q}_{\ell}$ of dimension 1. This induces an ℓ -adic representation $\rho_{\ell} : G_{\mathbb{Q}} \rightarrow \text{GL}(1, V_{\ell}(\mu)) \cong \mathbb{Q}_{\ell}^{\times}$ where for $\sigma \in G_{\mathbb{Q}}$, $\rho_{\ell}(\sigma)$ is determined by

$$\sigma(z) = z^{\rho_{\ell}(\sigma)} \text{ for } z \in \mu_{\ell^n}.$$

The above construction is known as the ℓ -adic cyclotomic character.

The motivation behind establishing some of these prerequisites of representations of G_K was to connect HGMs natural with these representations arising from point counts on generalised Legendre curves $C : y^N = x^i(1-x)^j(1-tx)^k$ and other more involved varieties. Constructive proofs

linking HGMs with these varieties can be found in [9, p. 6, 45-46], and further generalisations are also explored in [6, p. 2, 12-14]. Whilst of great interest and vital to the further study of HGMs, we refrain from giving an explicit construction here as such proofs are often lengthy and technically involved.

Finally, regarding the connection between classical and finite field hypergeometric functions, it is useful to view the latter as analogues of their analytic counterparts. Here, multiplicative characters χ_p play the role of the continuous parameter z in 5.1.7. Both the reflection formula and the fact that $B(x, y)$ can be expressed as a product of gamma functions further motivates their connection with Jacobi and Gauss sums respectively. Integral inductive formulas, such as those explored in [9, p. 18-19], demonstrate that higher order hypergeometric functions ${}_{n+1}F_n$ can be expressed in terms of integrals where lower-order ones appear

$${}_{n+1}F_n[\alpha_{n+1}; \beta_n; z] = \frac{1}{B(a_{n+1}, b_n - a_{n+1})} \int_0^1 f_{\alpha_{n+1}, \beta_n}(t) \cdot {}_nF_{n-1}[\alpha_n; \beta_{n-1}; zt] dt$$

where α_{n+1}, β_n are sets of complex parameters of size $n + 1$ and n respectively and $f_{\alpha_{n+1}, \beta_n}(t) = t^{a_{n+1}-1}(1-t)^{b_n-a_{n+1}-1}$. Just as the beta functions (and equivalently gamma functions) appear in the recursion of ${}_{n+1}F_n$, the same can be said about Jacobi and Gauss sums after comparing the notion of integration on $[0, 1]$ with summing over all $x \in \mathbb{F}_p$.

5.2. Jacobi sums, descent, and hypergeometric functions

We begin this section by refining Definition 5.1.9, identifying the terms in the sum with Grossencharacters coming from Jacobi sums. For the rest of this section, we set $\Psi_{\alpha, \beta}(\mathfrak{p})$ to be the Hecke character coming from the Jacobi sum $\tilde{J}_{\theta_{\alpha, \beta}}(\mathfrak{p})$ pre-descent, where $\theta_{\alpha, \beta} = \sum_{i=1}^n \{\alpha_i\} + \{-\beta_i\} - \{\alpha_i - \beta_i\}$. Due to 3.4.4 and [14, Theorem 4.3.4], we have that $\Psi_{\alpha+\mu/(q-1), \beta+\mu/(q-1)}(\mathfrak{p})$ also defines a Hecke character for all $\mu \in \{0, 1, \dots, q-2\}$ coming from its corresponding Jacobi sum. This allows us to introduce the simplified formula for $H_p(\alpha, \beta, t)$:

$$(5.2.1) \quad H_p(\alpha, \beta, t) = \frac{1}{q-1} \sum_{\mu=0}^{q-2} \Psi_{\alpha+\mu/(q-1), \beta+\mu/(q-1)}(\mathfrak{p}) \chi_p((-1)^n t)^{\mu/(q-1)}.$$

Thus, the HGM from before has been simplified to merely a sum of (twisted) Jacobi sums. One issue with working with these HGMs are the restrictions inherent to their construction. That is, the quantity $H_p(\alpha, \beta, t)$ is only defined for primes satisfying $\text{Nm}(\mathfrak{p}) \equiv 1 \pmod{m}$ (as per Definition

5.1.9), where m is the lowest common multiple of the denominators of the hypergeometric data α, β . For example, if we consider the case of $H_p(1/4, 1/2; 1, 1; t)$ then the only primes p for which $H_p(1/4, 1/2; 1, 1; t)$ is well-defined are those for which $4 \mid (\text{Nm}(p) - 1)$, or equivalently, $\text{Nm}(p) \equiv 1 \pmod{4}$.

Theorem 4.3.5 indicated that Jacobi sums descend to well-defined Grossencharacters on their field of moduli. In this chapter, we denote the descended Jacobi sum coming from $\Psi_{\alpha, \beta}(p)$ by $\psi_{\alpha, \beta}(p)$. Some of the work of getting around this restriction has already been done. In [3, Theorem 1.2], Beukers, Cohen, and Mellit generalise the point count of curves of the form $E_t(\mathbb{F}_q) : y^2 = x(x-1)(x-t)$ to be defined for all powers of primes q when the corresponding HGM is defined over $\mathbb{Q}$. By this, it is meant that, for $\alpha = \{\alpha_1, \dots, \alpha_n\}$ and $\beta = \{\beta_1, \dots, \beta_n\}$, $\prod_{i=1}^n (x - e^{2\pi i \alpha_i})$ and $\prod_{i=1}^n (x - e^{2\pi i \beta_i})$ are polynomials with integer coefficients. Obtaining a set of auxiliary data $p_1, \dots, p_r$ and $q_1, \dots, q_s$ by comparing the exponents in

$$\prod_{i=1}^n \frac{(x - e^{2\pi i \alpha_i})}{(x - e^{2\pi i \beta_i})} = \frac{\prod_{j=1}^r (x^{p_j} - 1)}{\prod_{j=1}^s (x^{q_j} - 1)}$$

an auxiliary HGM is able to be defined for all primes p , with the exception of those dividing the conductor of the Jacobi sum terms. We conclude this thesis by exploring how descent could apply to alleviating this restriction on the class of primes one can evaluate $H_p(\alpha, \beta, t)$ at.

Let $K = \mathbb{Q}(\zeta_m)$ and $K_{\alpha, \beta}$ the field of moduli of the Jacobi sum with defining data $\theta_{\alpha, \beta}$. Moreover, with $p \in \text{Spec}(\mathcal{O}_{K_{\alpha, \beta}})$ let $q = \text{Nm}(p)$ where q is the power of a prime number, the field of moduli $K_{\alpha, \beta, \mu}$ of $\psi_{\alpha + \mu/(q-1), \beta + \mu/(q-1)}$ satisfies $K_{\alpha, \beta} \subseteq K_{\alpha, \beta, \mu} \subseteq K_{\alpha, \beta}(\zeta_{q-1})$ for $\mu \in \{0, 1, \dots, q-2\}$. We denote by $F_{\alpha, \beta}$ the compositum of the fields $K_{\alpha, \beta, \mu}$, and say that $H_p(\alpha, \beta, t)$ descends to $F_{\alpha, \beta}$ if it can be defined as an HGM on this subfield.

Just as we saw in Example 4.3.7, the descent of Jacobi sums grants one the ability to input these Grossencharacters with a greater class of primes p (those satisfying $\text{Nm}(p) \not\equiv 1 \pmod{m}$). Thus, extending the result from Theorem 4.3.5 to HGMs should provide a path for defining $H_p(\alpha, \beta, t)$ on its field of moduli

$$H_p(\alpha, \beta, t) = \frac{1}{q-1} \sum_{\mu=1}^{q-2} \psi_{\alpha + \mu/(q-1), \beta + \mu/(q-1)}(p) \chi_p((-1)^n t)^{\mu/(q-1)}$$

with each $\Psi_{\alpha + \mu/(q-1), \beta + \mu/(q-1)}$ in (5.2.1) replaced with the descended Jacobi sum $\psi_{\alpha + \mu/(q-1), \beta + \mu/(q-1)}$. Unfortunately, computational restraints and the limited time available before finalising this project

prevented a complete proof of the descent of HGMs. Future work will therefore focus on formalising this descent process of HGMs, both algebraically and computationally, by constructing examples in Magma and exploring whether these descended HGMs preserve their intended purpose of counting points. The evidence above, however, strongly indicates that the descent of HGMs should be attainable.

References

- [1] Greg W Anderson. Cyclotomy and an extension of the taniyama group. *Compositio Mathematica*, 57(2):153–217, 1986. [1.2](#), [3.3](#)
- [2] George M Bergman. *An invitation to general algebra and universal constructions*. Springer, 2015. [2.1.2](#)
- [3] Frits Beukers, Henri Cohen, and Anton Mellit. Finite hypergeometric functions. *arXiv preprint arXiv:1505.02900*, 2015. [5.2](#)
- [4] Wieb Bosma, John Cannon, and Catherine Playoust. The magma algebra system i: The user language. *Journal of Symbolic Computation*, 24(3-4):235–265, 1997. [1.2](#)
- [5] Daniel Bump. *An introduction to the Langlands program*, volume 140. Springer Science & Business Media, 2003. [3.5](#)
- [6] Alyson Deines, Jenny G. Fuselier, Ling Long, Holly Swisher, and Fang-Ting Tu. Hypergeometric series, truncated hypergeometric series, and gaussian hypergeometric functions, 2015. [5.1](#)
- [7] Alyson Deines, Jenny G. Fuselier, Ling Long, Holly Swisher, and Fang-Ting Tu. Generalized legendre curves and quaternionic multiplication. *Journal of Number Theory*, 161:175–203, April 2016. [5.1](#)
- [8] Charles F. Doran, Tyler L. Kelly, Adriana Salerno, Steven Sperber, John Voight, and Ursula Whitcher. Hypergeometric decomposition of symmetric k3 quartic pencils, 2020. [5.1](#)
- [9] Jenny Fuselier, Ling Long, Ravi Ramakrishna, Holly Swisher, and Fang-Ting Tu. Hypergeometric functions over finite fields. *Mem. Amer. Math. Soc.*, 280(1382):vii+124, 2022. [3.1](#), [3.1](#), [5.1](#), [5.1.8](#), [5.1](#)
- [10] Carl Friedrich Gauss. *Disquisitiones Arithmeticae*. Springer New York, NY, 1986. [1.1](#)
- [11] John Greene. Hypergeometric functions over finite fields. *Transactions of the American Mathematical Society*, 301(1):77–101, 1987. [1.3](#)

- [12] Edmund Hlawka, Johannes Schoissengeier, and Rudolf Taschner. *Geometric and analytic number theory*. Springer Science & Business Media, 2012. [3](#)
- [13] Kenneth Ireland and Michael Ira Rosen. *A classical introduction to modern number theory*, volume 84. Springer Science & Business Media, 1990. [1.1](#), [3.1](#), [3.1.10](#), [3.3](#), [3.4.2](#), [3.5](#), [3.5](#), [3.5](#), [3.5](#)
- [14] Tyler L Kelly and John Voight. Hypergeometric motives from euler integral representations. *arXiv preprint arXiv:2412.03257*, 2024. [1.3](#), [3.1](#), [5.1](#), [5.1](#), [5.2](#)
- [15] Neal Koblitz. *p-adic Numbers, p-adic Analysis, and Zeta-Functions*, volume 58. Springer Science & Business Media, 2012. [2.1.2](#)
- [16] Leopold Kronecker. Zwei sätze über gleichungen mit ganzzahligen coefficienten. 1857. [3.4.2](#)
- [17] James S Milne. Fields and galois theory. *Courses Notes, Version*, 4, 2003. [5.1](#)
- [18] James S Milne. Class field theory, 2011. [4.4](#)
- [19] James S Milne. Lectures on étale cohomology, 2012. [2.2](#)
- [20] James S Milne. Motives—grothendieck’s dream. *Open problems and surveys of contemporary mathematics*, 6:325–342, 2013. [2.2](#)
- [21] James S Milne. Algebraic number theory, 2020. [2.1.1](#), [2.1.2](#), [2.1.2](#)
- [22] Pascal Molin and Aurel Page. Computing groups of hecke characters. *Research in Number Theory*, 8(4):91, 2022. [4.4](#)
- [23] Jürgen Neukirch. *Algebraic number theory*, volume 322. Springer Science & Business Media, 2013. [5.1](#)
- [24] David E Rohrlich. Jacobi sums and explicit reciprocity laws. *Compositio Mathematica*, 60(1):97–114, 1986. [3.4.2](#)
- [25] Norbert Schappacher. *Periods of Hecke characters*, volume 1301. Springer, 2006. [3.3](#), [3.3](#)
- [26] Jean-Pierre Serre. Motifs. In Lachaud Gilles, editor, *Journées arithmétiques de Luminy 17-21 Juillet 1989*, number 198-199-200 in Astérisque, pages 333–349. Société mathématique de France, 1991. [2.2](#), [2.2](#)
- [27] John Voight. *Quaternion algebras*. Springer Nature, 2021. [2.1.2](#), [4.4](#)
- [28] Lawrence C Washington. *Introduction to cyclotomic fields*, volume 83. Springer Science & Business Media, 2012. [3.4](#), [3.4.1](#), [3.4.2](#)

- [29] Mark Watkins. Computing with hecke grössencharacters. *Publications mathématiques de Besançon. Algèbre et théorie des nombres*, pages 119–135, 2011. [4.4](#)
- [30] Mark Watkins. Jacobi sums and grössencharacters. *Publications mathématiques de Besançon. Algèbre et théorie des nombres*, pages 111–122, 2018. [1.2](#), [1.2](#), [2.2](#), [3.3](#), [4.2](#), [4.2.13](#)
- [31] André Weil. Jacobi sums as “grössencharaktere”. In *Œuvres Scientifiques*, pages 63–71. Springer, 1952. [1.2](#), [3](#), [3.1](#), [3.4.2](#), [3.4.2](#)
- [32] Jared Weinstein. Reciprocity laws and galois representations: recent breakthroughs. *Bulletin of the American Mathematical Society*, 53(1):1–39, 2016. [1.1](#)